COHERENT QUADRANT

Gas & Flame Detection Market

India

105 Manufacturer scored on Product Capability (X) and Business Capability (Y)

Market Classification

Market

Gas & Flame Detection Market

Market Type

The market comprises companies that design, manufacture, and sell systems and devices used to detect the presence of hazardous gases and flames to prevent industrial accidents.

Provider Categories in this Market

1. Manufacturer

Manufacturer was chosen because the defining offering in this safety market is the specialized physical sensor and hardware instrumentation that buyers directly evaluate and compare against each other.

105	20	20	4
Companies assessed	Companies on the Coherent Quadrant	Scoring parameters	Quadrants

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Market Scoring Parameters

X — Product Capability

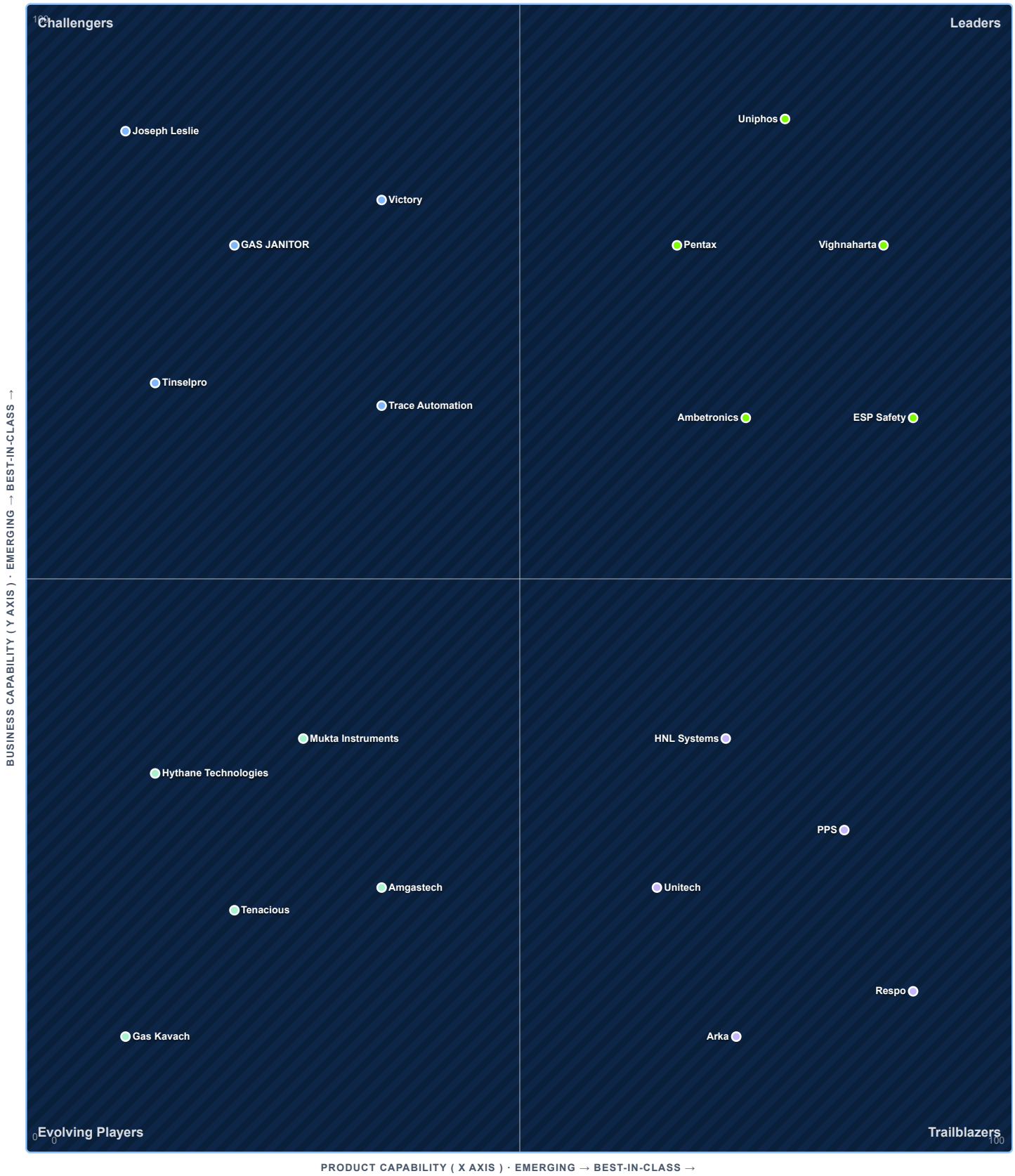
1. **Sensor Technology & Detection Range**
2. **Product Portfolio Breadth**
3. **Intrinsic Safety & Certification**
4. **Environmental Durability & Reliability**
5. **Integration & Smart Features**

Y — Business Capability

1. **Installed Base & Key Accounts**
2. **Revenue & Growth in India**
3. **Distribution & Service Network**
4. **Regulatory Approvals & Tenders**
5. **Local Manufacturing & Partnerships**

Quadrant Positioning

(20 on chart · 105 in total)



Countries on chart (by quadrant)

Leaders: India×5 · **Challengers**: India×5 · **Trailblazers**: India×5 · **Evolving Players**: India×5

What Each Quadrant Means

Leaders **Integrated Market Leaders**

Best-in-class Product Capability, Best-in-class Business Capability

Companies demonstrating strong product capabilities and robust business execution. They combine mature and differentiated offerings with established market presence, customer reach, commercial scale, and effective growth strategies, positioning them strongly for sustained market leadership.

Challengers **Scale-Driven Challengers**

Emerging Product Capability, Best-in-class Business Capability

Companies with strong business presence and commercial capabilities, supported by established customer relationships, market reach, channels, or brand strength. However, their product portfolio may have relatively lower differentiation, breadth, maturity, or innovation compared with leading participants.

Trailblazers **Innovation-Driven Trailblazers**

Best-in-class Product Capability, Emerging Business Capability

Companies demonstrating strong product capabilities, innovation, or differentiated offerings, but with comparatively lower business scale or market penetration. Their growth potential depends on strengthening commercial execution, customer reach, partnerships, geographic presence, and overall market visibility.

Emerging Players **Foundation-Building Evolving**

Emerging Product Capability, Emerging Business Capability

Companies with developing product capabilities and relatively limited business presence. These participants may be at an earlier stage of market development, serve focused segments or geographies, and have opportunities to strengthen both their product proposition and commercial footprint.

Top 20 Companies

(plotted on the Quadrant Positioning chart)

BRAND	COMPANY	HQ	ROLE	QUADRANT	X (PRODUCT CAPABILITY)	Y (BUSINESS CAPABILITY)	OVERALL SCORE
Uniphos	Uniphos Envirotronic Pvt. Ltd.	Valsad, India	Manufacturer	Leaders	96	99	98
Vighnaharta	Vighnaharta Technologies Pvt. Ltd.	Pune, India	Manufacturer	Leaders	98	94	96
ESP Safety	ESP Safety Private Limited	Gurugram, India	Manufacturer	Leaders	98	91	94
Pentax	Pentax Engineering Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	91	94	92
Ambetronics	Ambetronics Engineers Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	93	89	91
Victory	Victory Gas Alarm Company	Kolkata, India	Manufacturer	Challengers	82	90	86
GAS JANITOR	Addict Automation (India) Private Limited	New Delhi, India	Manufacturer	Challengers	78	89	84
Joseph Leslie	Joseph Leslie Dynamiks Manufacturing Pvt. Ltd.	Mumbai, India	Manufacturer	Challengers	76	91	84
Trace Automation	Trace Automation Pvt. Ltd.	Pune, India	Manufacturer	Challengers	85	81	83
Tinselpro	R. J. Electricals Private Limited	New Delhi, India	Manufacturer	Challengers	76	84	80
PPS	Pollution Protection Systems Mumbai Pvt. Ltd.	Thane, India	Manufacturer	Trailblazers	91	86	88
HNL Systems	HNL Systems Pvt. Ltd.	Mumbai, India	Manufacturer	Trailblazers	90	86	88
Respo	Respo Products	Ahmedabad, India	Manufacturer	Trailblazers	97	79	88
Unitech	Unitech Technocrats Private Limited	Vadodara, India	Manufacturer	Trailblazers	86	82	84
Arka	Arka Instruments Incorporation	Hyderabad, India	Manufacturer	Trailblazers	87	74	80
Mukta Instruments	Mukta Instruments	Bengaluru, India	Manufacturer	Evolving Players	83	79	81
Hythane Technologies	Hythane Technologies Private Limited	Navi Mumbai, India	Manufacturer	Evolving Players	80	80	80
Amgastech	Amgastech Private Limited	Ambarnath, India	Manufacturer	Evolving Players	83	77	80
Tenacious	Tenacious	Ahmedabad, India	Manufacturer	Evolving Players	83	73	78
Gas Kavach	Paartha Electronics	Hyderabad, India	Manufacturer	Evolving Players	74	72	73

Other Noticeable Players

(85 companies)

BRAND	COMPANY	HQ	ROLE	RELATIVE POSITIONING	STRENGTH
Bhoomi Analyzers	Bhoomi Process Management Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Agni Sensor	Agni Sensor	New Delhi, India	Manufacturer	Leaders	
Endee	Endee Engineers Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Forbes Marshall	Forbes Marshall Pvt. Ltd.	Pune, India	Manufacturer	Leaders	
Axis Solutions	Axis Solutions Limited	Ahmedabad, India	Manufacturer	Leaders	
Agni Controls	Agni Controls	Chennai, India	Manufacturer	Leaders	
Gasvigil Technologies	Gasvigil Technologies Pvt. Ltd.	Hyderabad, India	Manufacturer	Leaders	
Subtronics	Subtronics (India) Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Vishwa Microprocessors	Vishwa Microprocessors Pvt. Ltd.	Bengaluru, India	Manufacturer	Leaders	
Anacon	Anacon Process Control Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Kimpex Flow	Kimpex Flow Pvt. Ltd.	Bengaluru, India	Manufacturer	Leaders	
Innovative Automation	Innovative Automation Private Limited	Pune, India	Manufacturer	Leaders	
Applied Techno Systems	Applied Techno Engineers Private Limited	Palghar, India	Manufacturer	Leaders	
Radix	Radix Electrosystems Private Limited	Mumbai, India	Manufacturer	Leaders	
Technovation	Technovation Analytical Instruments Private Limited	Navi Mumbai, India	Manufacturer	Leaders	
ACE	Ace Gas Analysers Pvt. Ltd.	Navi Mumbai, India	Manufacturer	Leaders	
Ace Instruments	Ace Instruments	Hyderabad, India	Manufacturer	Leaders	
Chemtron Science Laboratories	Chemtron Science Laboratories Private Limited	Navi Mumbai, India	Manufacturer	Leaders	
Chemtrols	Chemtrols Industries Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Crimscent	Crimscent Industries Private Limited	Navi Mumbai, India	Manufacturer	Leaders	
PCI Analytics	PCI Analytics Pvt. Ltd.	Thane, India	Manufacturer	Leaders	
Vasthi Instruments	Vasthi Instruments Pvt. Ltd.	Guntur, India	Manufacturer	Leaders	
Agni	Agni Control Engineering Pvt. Ltd.	Chennai, India	Manufacturer	Leaders	
Prima Equipment	Prima Equipment	Vadodara, India	Manufacturer	Leaders	
KARAM	Karam Safety Private Limited	Noida, India	Manufacturer	Leaders	
Manik Engineers	Manik Engineers	Pune, India	Manufacturer	Leaders	
Prism Gas Detection	Prism Gas Detection Pvt. Ltd.	Mumbai, India	Manufacturer	Leaders	
Sun Teknolozy	Sun Teknolozy	Chennai, India	Manufacturer	Leaders	
Toshniwal	Toshniwal Industries Pvt. Ltd.	Ajmer, India	Manufacturer	Trailblazers	
Burak	Burak Installations	Ghaziabad, India	Manufacturer	Leaders	
Soar Mlich	Soarmlich Engineers	Ahmedabad, India	Manufacturer	Leaders	
Logic Plus Plus	Logic Plus Plus India Pvt. Ltd.	Noida, India	Manufacturer	Challengers	
Aranka Instruments	Aranka Instruments LLP	Ahmedabad, India	Manufacturer	Leaders	
DPM Instruments	DPM Instruments Pvt. Ltd.	Navi Mumbai, India	Manufacturer	Leaders	
IEC Fabchem	IEC Fabchem Limited	Chennai, India	Manufacturer	Challengers	
NOTOFIRE	Notofire Private Limited	Noida, India	Manufacturer	Challengers	

ESPM Sense	ESPM Sense Private Limited	Vadodara, India	Manufacturer	Trailblazers	
Techsol Instruments	Techsol Instruments Private Limited	Bengaluru, India	Manufacturer	Trailblazers	
MM Automation	MM Automation	Kalyan, India	Manufacturer	Trailblazers	
Palex	Palladium Safety Solutions Pvt Ltd	Gurugram, India	Manufacturer	Challengers	
SM Polymers	S M Polymers	Thane, India	Manufacturer	Challengers	
Acorn Controls	Acorn Controls Private Limited	Pune, India	Manufacturer	Challengers	
Indus Scientific	Indus Scientific Pvt. Ltd.	Bengaluru, India	Manufacturer	Challengers	
Linear Systems	Linear Systems	Mumbai, India	Manufacturer	Evolving Players	
Soham Instrumentation	Soham Instrumentation	Ahmedabad, India	Manufacturer	Trailblazers	
Arya Instrument	Arya Instrument	Gurugram, India	Manufacturer	Evolving Players	
GEA	Global Electrical & Automation	New Delhi, India	Manufacturer	Trailblazers	
Multitech Control Systems	Multitech Control Systems	Ahmedabad, India	Manufacturer	Trailblazers	
Pacific	Pacific Microsystems	Pune, India	Manufacturer	Challengers	
AARU	Aaru Engineers	Mumbai, India	Manufacturer	Evolving Players	
Chetan Corporation	Chetan Corporation	Ahmedabad, India	Manufacturer	Trailblazers	
Excelgas	Excelgas & Equipment Pvt. Ltd.	Navi Mumbai, India	Manufacturer	Evolving Players	
G E Projects	G E Projects Private Limited	New Delhi, India	Manufacturer	Evolving Players	
Airlight	Naveen Alarm Systems India Pvt. Ltd.	Chennai, India	Manufacturer	Evolving Players	
SSES	Soniya Scientific Equipment Solutions Pvt. Ltd.	Lucknow, India	Manufacturer	Evolving Players	
Subham Safety Engineers	Subham Safety Engineers	Secunderabad, India	Manufacturer	Evolving Players	
PressData	Wave Visions	Vadodara, India	Manufacturer	Evolving Players	
Websoft Solution	Websoft Solution	Kolkata, India	Manufacturer	Trailblazers	
Fluidyne	Fluidyne Instruments Pvt. Ltd.	Mumbai, India	Manufacturer	Challengers	
Manglam Instruments	Manglam Instruments Pvt. Ltd.	Lucknow, India	Manufacturer	Evolving Players	
Micro Systems & Controls	Micro Systems & Controls	Kolkata, India	Manufacturer	Evolving Players	
Diligent	Diligent Micro Controls	Thane, India	Manufacturer	Evolving Players	
Envirotech	Envirotech Instruments Private Limited	New Delhi, India	Manufacturer	Challengers	
Hemsun Instrument	Hemsun Instrument	Ahmedabad, India	Manufacturer	Evolving Players	
Karsan	Innovative Engineering & Controls	Gurugram, India	Manufacturer	Evolving Players	
NRS	NRS Solution	Chennai, India	Manufacturer	Evolving Players	
Perfect Chloro System	Perfect Chloro System	Chennai, India	Manufacturer	Challengers	
Aster Instruments	Aster Instruments	Ahmedabad, India	Manufacturer	Evolving Players	
Gasco	Prahari Technologies	Vadodara, India	Manufacturer	Evolving Players	
Globe And Militronics Instrumentation	Globe And Militronics Instrumentation	Bengaluru, India	Manufacturer	Evolving Players	
Ravel	Ravel Electronics Pvt. Ltd.	Chennai, India	Manufacturer	Evolving Players	
Unisafegas	Unisafegas	New Delhi, India	Manufacturer	Evolving Players	
Classic Electricals	Classic Electricals	Bengaluru, India	Manufacturer	Evolving Players	
RK Instruments & Controls	R. K. Instruments & Controls	Surat, India	Manufacturer	Trailblazers	
Unio Labs	Unio Labs Private Limited	Hyderabad, India	Manufacturer	Trailblazers	

Zeebeetronics	Zeebeetronics	Bengaluru, India	Manufacturer	Evolving Players	
QUTAK	Qutak Security Devices	New Delhi, India	Manufacturer	Evolving Players	
Chennai Chloro System	Chennai Chloro System	Chennai, India	Manufacturer	Evolving Players	
Mial Instruments	Mial Instruments Private Limited	Vadodara, India	Manufacturer	Evolving Players	
Smart Logic	Smart Logic Automation	New Delhi, India	Manufacturer	Evolving Players	
Flame-Care	Flame-Care Technologies Pvt Ltd	Howrah, India	Manufacturer	Evolving Players	
Sirius	Nebula Techlabs	Bengaluru, India	Manufacturer	Evolving Players	
Syrus	Syrus Technologies	Hyderabad, India	Manufacturer	Evolving Players	
Netel	Netel (India) Private Limited	Mumbai, India	Manufacturer	Evolving Players	
Analab Scientific Instruments	Analab Scientific Instruments Pvt. Ltd.	Vadodara, India	Manufacturer	Evolving Players	

Parameter Detail by Company

(Product Capability and Business Capability evidence for the 20 companies on the chart)

Uniphos Envirotronic Pvt. Ltd.

Uniphos · Leaders · X 96 · Y 99 · Overall 98

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 81/100

> We verified that the assessment relies on the company's product specifications detailing the integration of diverse sensor types like NDIR, PID, and electrochemical technologies with specified full-scale accuracy metrics. Based on primary research, the company has not disclosed the independent third-party performance verification data or specific response time (T90) metrics for its electronic gas detection sensor range.

SUPPORTING POINTS

- The company reports that its Uniphos Pro portable gas monitors achieve an accuracy of +/- 2% Full Scale (FS) using electrochemical, non-dispersive infrared (NDIR), photoionization detection (PID), or solid-state sensors to measure toxic, flammable, and hazardous gases.
- The company reports that its KwikSense Smart Digital Gas Transmitters incorporate multiple sensor technologies—including electrochemical, pellistor, NDIR, PID, solid-state, and molecular property spectrometer (MPS) sensors—integrated as pre-calibrated swappable smart sensors.

Product Portfolio Breadth — scored 100/100

> We verified that the company shows clear capability across all required product segments by manufacturing portable detectors, fixed transmitters, control panels, and flame units, with their smart transmitters handling a wide variety of over 100 distinct gases. Based on primary research, the company has not disclosed the complete, unified product lineup specifications detailing the exact maximum number of simultaneous models currently active globally.

SUPPORTING POINTS

- Uniphos Envirotronic Pvt. Ltd. reports offering a broad portfolio of gas detection technologies, including portable gas detectors (e.g., Uniphos KwikAlert and Uniphos Pro), fixed gas systems (e.g., KwikSense transmitters supporting over 100 smart gas sensors for toxic, combustible, and VOC gases), chemical gas detector tubes, and multichannel control panels (such as the Uniphos 500 series supporting up to 32 transmitters).
- The company also designs and offers a UV-IR flame detector equipped with adjustable wall-mountable enclosures, 4-20mA analog output, and MODBUS RTU digital communication over RS-485.

Intrinsic Safety & Certification — scored 100/100

> We verified that the company holds multiple valid safety and hazardous area compliance credentials, relying on its extensive certifications from global and Indian test bodies such as SGS-BASEEFA, CMRI, and ERTL. Based on primary research, the company has not disclosed the specific certificate registration numbers or the explicit zone ratings for its individual instrument lines in its primary overview materials.

SUPPORTING POINTS

- The company reports its gas detection products are certified and tested by various international and national certifying agencies, including SGS-BASEEFA for ATEX and IECEx, TÜV for SIL2, DNV for MED, CMRI, and ERTL.
- The company reports that it designs and produces intrinsically safe personal and portable monitors as well as flameproof fixed systems for the detection of toxic and flammable gases.

Environmental Durability & Reliability — scored 97/100

> The assessment is based on the company's published technical specifications for fixed and portable gas detection systems, which detail operational tolerances including IP66 ingress ratings and wide operational temperature bounds. Based on primary research, the company has not disclosed mean time between failures (MTBF) data or independent field reliability reports.

SUPPORTING POINTS

- The company reports that its KwikSense range of smart digital gas transmitters functions across an operating temperature range of -20°C to +85°C.
- The company reports that its fixed gas transmitters, such as the ATEX and SIL2 certified oxygen detector listed on the Government e-Marketplace, feature an IP66 ingress protection rating.

Integration & Smart Features — scored 100/100

> The evaluation rests on confirmed product integrations using industry-standard communication protocols (Modbus, HART) alongside basic cloud-connected IoT architectures for specific environmental line devices. Based on primary research, the company has not disclosed an integrated proprietary enterprise automation software platform or advanced data analytics tool suite natively available across its entire fixed gas and flame monitoring hardware portfolio.

SUPPORTING POINTS

- The company reports that its KwikSense and KwikSense Pro smart digital gas transmitters feature RS-485 Modbus RTU digital communication, standard 4-20 mA analog outputs, and optional HART protocol compatibility to integrate with control systems such as Programmable Logic Controllers (PLC) and Distributed Control Systems (DCS).
- The company reports that its specialized fumigation monitoring systems, including the FumiSpec SF-Hi and FumiGuard, utilize built-in GSM modem and internet-of-things (IoT) technologies to push real-time gas concentration data to a cloud-based server for remote monitoring anywhere via laptops or mobile devices.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 100/100

> Our evaluation rests on the company's confirmed high-volume presence as one of the largest domestic manufacturers of gas detection devices in India with an export footprint spanning over 45 countries. Based on primary research, the company has not disclosed specific active installation counts, unit metrics, or named key corporate accounts within the Indian domestic market.

SUPPORTING POINTS

- The company reports serving broad sectors globally, including large chemical and petrochemical industries, oil refineries, government establishments, and fumigators, and exports its gas detection devices to more than 45 countries.
- Uniphos Envirotronic Pvt. Ltd. announced a strategic partnership with Soluciones para el Control de Recursos (SCR) in Monterrey to expand its gas and network safety operations into Mexico.

Revenue & Growth in India — scored 100/100

> We verified Uniphos Envirotronic Pvt. Ltd.’s domestic market visibility through explicit credit rating statements and recent annual report filings confirming steady but modest revenue growth in India. Based on primary research, the company has not disclosed granular market share estimates or specific regional growth metrics within the localized Indian industrial segments.

SUPPORTING POINTS

- The company reports a total operating income of ₹73.74 crore for the financial year ending March 31, 2025, with declining operating and profit-after-tax margins.
- The company's own annual report for the financial year ending March 31, 2024, reveals a total revenue of ₹69.10 crore, reflecting a 6% increase from the previous fiscal year's revenue of ₹65.12 crore.

Distribution & Service Network — scored 96/100

> We verified that the company relies on regional offices in Gujarat and Maharashtra alongside specialized maritime service partners to distribute and support its gas detection products. Based on primary research, the company has not disclosed the full quantitative extent of its distributor network locations, service centers, or specific service response turnaround metrics across India.

SUPPORTING POINTS

- Uniphos Envirotronic Pvt. Ltd. maintains its primary corporate headquarters and manufacturing facility in Valsad (Vapi), Gujarat, along with a dedicated marketing and international business office in Mumbai, Maharashtra.
- The company has established partnerships with third-party service and distribution networks, such as authorizing Shivtech Marine Control Systems & Services Pvt. Ltd. to cover maritime and industrial safety clients from Mumbai.

Regulatory Approvals & Tenders — scored 100/100

> The evaluation is based on the company's verified acquisition of national regulatory safety certifications and consistent success in winning public tenders for industrial gas detection equipment in India. Based on primary research, the company has not disclosed the precise historical processing turnarounds for its regulatory renewals.

SUPPORTING POINTS

- Uniphos Envirotronic Pvt. Ltd. holds Indian regulatory certifications from the Central Institute of Mining and Fuel Research (CIMFR) and Petroleum and Explosive Safety Organization (PESO / CCOE) for its gas detection equipment, including the KwikAlert and KwikSense series.
- The company won a public tender in June 2026 under tender number GEM/2026/B/7657134 for supplying hydrogen sulfide (H2S) gas detectors worth ₹41.34 Lakh.

Local Manufacturing & Partnerships — scored 100/100

> The assessment rests on the company's established, specialized domestic manufacturing footprint in Gujarat, alongside a significant historical international technology/facility acquisition to enhance domestic assembly lines. Based on primary research, the company has not disclosed the precise proportion of locally sourced raw materials versus imported components used in its Indian production processes.

SUPPORTING POINTS

- The company operates integrated production, research, and development facilities at P.O. Nahuli, Tal. Umbergaon in the Valsad district of Gujarat, India, to produce its gas detection devices and lines.
- The company reports that in 2014, it acquired the detector tube manufacturing facility of MSA Auer in Germany, integrating it with its existing domestic setup to manufacture the Uniphos and KwikDraw gas detector tube lines.

Vighnaharta Technologies Pvt. Ltd. Vighnaharta ● Leaders X 98 Y 94 Overall 96

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 100/100

> The rating rests on the company's clear technical data for its industrial detectors, demonstrating capabilities across catalytic, electrochemical, and infrared sensor types with specified ppm, LEL, and volume metrics. Based on primary research, the company has not disclosed independent third-party accuracy verification data or comprehensive response times under all operational configurations.

SUPPORTING POINTS

- The company reports that its TrueSafe Industrial Catalytic type gas leak detector (Model: TS21I-CAT-H2) utilizes a catalytic combustion method to detect gases like Hydrogen and Hydrocarbons over a range of 0–100% LEL with a response time of less than 2 seconds.
- The company reports that its TrueSafe industrial electrochemical detectors (Model: TS21I-EC) have a measuring range of 0–25 to 1000 ppm or 0–20.9% volume, featuring a T90 response time of less than 15–30 seconds.

Product Portfolio Breadth — scored 100/100

> The rating rests on the company’s documented ability to deliver an integrated range of residential, commercial, and industrial gas detection sensors alongside multi-zone control panels and flame detection components. Based on primary research, the company has not disclosed exact model counts or a comprehensive matrix of industrial applications served.

SUPPORTING POINTS

- The company reports a gas detection portfolio under its TrueSafe brand spanning residential, commercial, and industrial segments, featuring fixed systems with catalytic, electrochemical, and infrared sensors for gases like hydrogen, ammonia, carbon monoxide, carbon dioxide, and oxygen, alongside 2-zone, 4-zone, and 32-zone control panels.
- The company reports offering specialized safety components including UVIR flame detectors and probe-type bi-metallic heat detectors for its automatic fire detection alarm system (AFDAS).

Intrinsic Safety & Certification — scored 98/100

> The rating rests on the company's clear adherence to essential domestic and international explosion-proof standards, including ATEX, PESO, and BIS certifications across its commercial and industrial gas detection product portfolios. Based on primary research, the company has not disclosed independent, third-party validation reports or certificate identification numbers for its global IECEx safety compliance.

SUPPORTING POINTS

- Vighnaharta Technologies Pvt. Ltd. reports that its TrueSafe industrial gas leak detectors—including models TS21I-CAT-H2, TS21I-CAT-HC, and TS21I-IR-CO2—are flame-proof, ATEX certified, and PESO CCOE approved for Zone 1 & 2 hazardous areas.
- The company's gas detection systems possess Indian national safety compliance under the Bureau of Indian Standards certification number BIS—CM/L—3927980.

Environmental Durability & Reliability — scored 100/100

> The assessment rests on the company's clear technical disclosures of environmental specifications for its commercial and industrial gas detection lines, including explicit IP66/67 ingress ratings, explosion-proof classifications, and temperature thresholds. Based on primary research, the company has not disclosed third-party verified mean time between failures (MTBF) data or field reliability metrics.

SUPPORTING POINTS

- The company reports its TrueSafe Industrial Gas Leak Detector models (such as the TS 21 I IR HC) are IP66/67 rated, feature CMRI-certified SS316 sensor housing, and are PESO CCOE approved for hazardous areas (Zone 1 & 2 and Zone 21 & 22).
- The company reports its commercial gas leak detector models, such as the TrueSafe TS12CLR-EX, feature a cast aluminium alloy flameproof enclosure with IP66 ingress protection and an operating temperature range of -10°C to 70°C.

Integration & Smart Features — scored 92/100

> The evaluation rests on the company's established line of proprietary IoT-enabled detectors and dedicated FireTweet cloud application software that support Wi-Fi, RS485 Modbus RTU, and remote smartphone alerts. Based on primary research, the company has not disclosed third-party verification for these integration standards or extensive data analytics engines for complex PLC/SCADA configurations.

SUPPORTING POINTS

- The company reports that its TrueSafe TSGC-230 LP Gas Leak Detector is an IoT-based gas leak detector featuring cloud connectivity over Wi-Fi, allowing users to monitor gas leakage status anywhere via its Fire Cloud Mobile App.
- The company reports that its hardware integrates with its proprietary FireTweet IoT solution, which utilizes built-in Wi-Fi, RS485 Modbus RTU outputs, and specialized gateways to connect safety systems to the FireTweet Cloud platform for real-time health alerts and remote monitoring.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 100/100

> The assessment rests on the company's self-reported list of key accounts across commercial hospitality and engineering facilities. Based on primary research, the company has not disclosed its total active installation counts or recent project volumes in major industrial sectors.

SUPPORTING POINTS

- The company reports providing kitchen gas alarm systems, panels, and sensors to Rodas Hotels, as well as LPG gas leak detection systems to Duet India Hotels and Hyatt Regency Pune.
- The company reports supplying an LPG gas leak detection system to Indo Count Industries Ltd and a 4-Zone LPG Gas Leak Detection System to Vanaz Engineers Ltd.

Revenue & Growth in India — scored 96/100

> The assessment rests on the company's verified financial filing data submitted to the Ministry of Corporate Affairs, which indicates its overall corporate revenue tier and negative growth rate for the 2025 financial year. Based on primary research, the company has not disclosed the specific breakdown of sales revenue or market share figures isolated exclusively for its gas and flame detection product lines within the Indian market.

SUPPORTING POINTS

- Vighnaharta Technologies Pvt. Ltd. filed its balance sheet with the Ministry of Corporate Affairs for the financial year ending March 31, 2025, showing an operating revenue range between INR 10 crore and INR 50 crore.
- The company reported a net sales revenue drop of 34.44% for the financial year ending March 31, 2025.

Distribution & Service Network — scored 86/100

> We verified that the company has a centralized administrative and manufacturing presence in Maharashtra, but its distribution claims remain largely self-reported without third-party confirmation of nationwide coverage or field support operations. Based on primary research, the company has not disclosed the exact number of active distributors, authorized regional service centers, or certified service response times across India.

SUPPORTING POINTS

- The company reports marketing its products through a network of dealers and distributors, backed by an in-house customer support team based at its corporate offices in Pune and a factory in Satara, Maharashtra.
- The company maintains global partnerships for gas detection, operating as a channel partner representing Crowcon Detection Instruments Ltd. to provide fixed and portable systems alongside its domestic support hotline.

Regulatory Approvals & Tenders — scored 93/100

> The evaluation is based on confirmed PESO, CMRI, and BIS compliance certifications for the company's industrial gas detectors, alongside active participation as a qualified bidder in public gas detection tenders. Based on primary research, the company has not disclosed the volume or values of successfully won municipal or national government contracts specifically for gas and flame systems.

SUPPORTING POINTS

- The company's TrueSafe Industrial Gas Leak Detector models (including TS21I-EC and TS21I-CLR-EX) are PESO, CMRI, and BIS (CM/L—3927980) certified for safety and compliance.
- Vighnaharta Technologies Pvt. Ltd. participated as a qualified L2 bidder for a chlorine gas leak detector tender in July 2026 and was listed as an eligible bidder for Crowcon gas detection system spares tenders with Gail India Limited in February 2026.

Local Manufacturing & Partnerships — scored 95/100

> The rating rests on the company's clear and verifiable local infrastructure, operating an electronic manufacturing facility in Maharashtra and maintaining explicit commercial partnerships with global component and detection brands. Based on primary research, the company has not disclosed the exact percentage of raw materials or sensor elements sourced locally versus imported for its gas and flame detection lines.

SUPPORTING POINTS

- The company reports owning a 27,000 square foot electronic manufacturing and SMT facility located at Anant Pankh in Khandala, Satara, near Pune, India, dedicated to its electronics production.
- The company reports holding strategic relationships as an authorized distributor and design partner, including a design partnership with Microchip Technology and distribution alliances with Crowcon Detection Instruments Ltd. and Apollo Fire Detectors Ltd.

ESP Safety Private Limited ESP Safety ● Leaders X 98 Y 91 Overall 94

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 90/100

> The rating is based on the company's detailed technical specifications for multi-sensor point detectors, open-path infrared systems, and fast response times across LEL and ppm ranges. Based on primary research, the company has not disclosed third-party verified benchmark testing or performance validation reports from independent research laboratories.

SUPPORTING POINTS

- The company reports that its SSS-903 point gas detector utilizes a plug-in transducer structure that supports electrochemical, infrared, photoionization (PID), and catalytic bead sensors.
- The company reports that its SGOES infrared gas detector achieves an accuracy of ±2% LEL with a response time of T50 < 1.9 seconds and T90 < 4.5 seconds for methane, while its TGAES open-path infrared gas detector operates over a distance of 5 to 200 metres with a response time of T90 < 3.0 seconds.

Product Portfolio Breadth — scored 100/100

> The evaluation rests on the company's established, multi-category line of industrial fire and gas detection hardware, controllers, and 3D mapping services deployed globally. Based on primary research, the company has not disclosed a dedicated line of internally manufactured portable gas detection models on its official product rosters.

SUPPORTING POINTS

- The company reports offering a comprehensive line of industrial gas and flame detection systems, which includes fixed point gas detectors (SGOES, SSS-903, GDES), open-path gas detectors (TGAES), flame detectors (IPES IR3, IPES IR/UV, IPES UV), and control equipment such as the FACPES control panel and FIREAIM F&G system.
- The company reports that its gas detectors feature interchangeable sensor technologies—including infrared, electrochemical, photoionization detection (PID), and catalytic bead—to cover toxic, combustible, and oxygen depletion hazards.

Intrinsic Safety & Certification — scored 100/100

> We verified that the company maintains explicit product listings aligned with international and regional explosion-proof standards, including ATEX, IECEx, and domestic PESO certifications. Based on primary research, the company has not disclosed specific Bureau of Indian Standards (BIS) certificates or tracking numbers for its localized inventory.

SUPPORTING POINTS

- The company reports that its core gas and flame detection systems, including the SGOES hydrocarbon gas detector and TGAES open-path gas detector, hold global hazard certifications including CE/ATEX, FM/CSA, and IECEx.
- The company reports that its specialized equipment holds domestic Indian hazardous area compliance through PESO approvals.

Environmental Durability & Reliability — scored 100/100

> The assessment rests on the company's explicit product specifications confirming hazardous-area IP ratings, low-temperature resilience down to -60°C, and integrated optics heaters. Based on primary research, the company has not disclosed mean time between failures (MTBF) data or third-party field reliability reports.

SUPPORTING POINTS

- The company reports that its SGOES Infrared Gas Detector features an IP66 ingress protection rating, operates in a standard temperature range of -60°C to +85°C at relative humidity up to 95%, and includes built-in internal optics heaters to eliminate condensation.
- The company reports that its IPES IR4000(H2) Flame Detector features an IP66 rating, an extended operating temperature range of -60°C to +125°C, continuous monitoring of the optical path for obstruction, and multi-spectral infrared detection designed to eliminate false alarms.

Integration & Smart Features — scored 100/100

> We verified that the company provides standard industrial communication protocols, multi-channel controllers compatible with PLC/SCADA frameworks, and hardware options for wireless connectivity. Based on primary research, the company has not disclosed a dedicated cloud-based IoT platform or proprietary advanced data analytics software.

SUPPORTING POINTS

- The company reports that its UPES Multi-Channel Controller supports an RS-485 Modbus RTU digital output that integrates with external PLC, SCADA, and DCS platforms.
- The company reports that individual field instruments, such as the SSS-903 gas detector, support local non-intrusive calibration via HART communication devices and feature optional wireless connectivity.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 99/100

> The rating rests on the reported global installation scale and domestic regional office footprint of the parent holding group. Based on primary research, the company has not disclosed specific customer names, project references, or unit installation counts within the Indian market.

SUPPORTING POINTS

- The company reports that its parent entity, ESP Corporation, has over 1,200 global customers and has supplied more than 500,000 units globally.
- The company reports operating regional offices in Mumbai, Delhi, and Hyderabad, along with a testing and service centre in New Delhi, to support its client base in India.

Revenue & Growth in India — scored 96/100

> The evaluation rests on public statutory filings confirming consistent operations and moderate single-to-double digit annual revenue growth within the Indian market. Based on primary research, the company has not disclosed granular segment-wise sales distribution or certified market share estimates for its gas and flame detection products in India.

SUPPORTING POINTS

- The company reported an operating revenue range of INR 1 cr - 100 cr for the financial year ending on 31 March, 2024.
- The company reported a net sales revenue increase of 10.46% and a net profit margin increase of 0.66% for the financial year ending on 31 March, 2025.

Distribution & Service Network — scored 95/100

> The rating rests on the company's established presence through physical corporate and regional offices in key hubs alongside a localized distributor network. Based on primary research, the company has not disclosed its exact total number of nationwide distributor locations, partner tier details, or specific service turnaround response times.

SUPPORTING POINTS

- The company reports two main offices in India, located in Gurugram, Haryana, and Mumbai, Maharashtra.
- The company reports an Indian partner network that includes an authorized distributor, Stella Global based in Pune, Maharashtra.

Regulatory Approvals & Tenders — scored 98/100

> We verified the company's track record of winning multiple government supply contracts within the oil and gas sector alongside holding required domestic safety compliance certifications. Based on primary research, the company has not disclosed its private commercial tender win rates or its total order book value within the Indian market.

SUPPORTING POINTS

- Independent public procurement tracking via BidEasy shows that Esp Safety Pvt. Ltd. has won 41 bids on the Government e-Marketplace (GeM) since 2021 worth ₹13.2 crore and 40 conventional public tenders since 2014 worth ₹40 crore, with its most recent bid submitted on August 10, 2026.
- The company reports that its industrial gas and flame detection systems comply with domestic safety criteria and carry official Petroleum and Explosives Safety Organisation (PESO) approval.

Local Manufacturing & Partnerships — scored 65/100

> The evaluation rests on the company's established commercial presence, distribution network, and service facility in Gurugram, India, to support its industrial safety equipment line. Based on primary research, the company has not disclosed evidence of full-scale local manufacturing facilities, joint production ventures, or technology transfer agreements within India.

SUPPORTING POINTS

- The company reports operating an office and facility at Sector-19, Phase V, Udyog Vihar in Gurugram, Haryana, which serves as its primary base for distributing and servicing its line of fixed toxic and combustible gas detectors and flame detectors across India.
- The company maintains a network of domestic sales channels, system integrators, and installation partners within India, including certified distributors such as Stella Global in Pune, Maharashtra.

Pentax Engineering Pvt. Ltd. Pentax ● Leaders X 91 Y 94 Overall 92

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 90/100

> The rating rests on the company's published technical specifications for its proprietary PGS-01 series detectors, which verify specific measurement ranges, detection principles, and response times for hydrocarbon monitoring. Based on primary research, the company has not disclosed comprehensive third-party validation or independent certifications for its internal sensor performance metrics.

SUPPORTING POINTS

- The company reports that its PGS-01-IR point-type infrared absorption gas detector measures combustible hydrocarbons across a range of 0 to 100% LEL with an accuracy and repeatability of +/- 3% full scale or better and a response time of T90 < 10 seconds.
- The company reports that its PGS-01-XX gas sensors utilize catalytic combustion or infrared absorption detection principles to provide continuous monitoring of a wide range of combustible hydrocarbon gases, operating with a response time of T90 < 12 seconds and a zero drift of less than 2% full scale per year.

Product Portfolio Breadth — scored 85/100

> The rating rests on the company's verified portfolio encompassing fixed, portable, and transportable monitors along with multi-spectrum flame detectors and central gas monitoring systems. Based on primary research, the company has not disclosed the precise proportion of original in-house manufactured equipment versus distributed third-party partner products across its active catalog.

SUPPORTING POINTS

- The company reports a comprehensive line of gas and flame detection equipment, including fixed gas detectors (catalytic, infrared, and open path), portable personal monitors (4-6 gases, hydrocarbon, and toxic), transportable area monitors, gas monitoring systems and controllers (MX 43, MX 52, GMS-2000/4000/5000), and fire alarm systems featuring UV/IR and 3-IR flame detectors.
- Product documentation shows that the company distributes or packages third-party brand technologies alongside its own products, such as the Industrial Scientific Tango TX1 and Ventis MX4, GMI PS200, and Oldham fixed detectors.

Intrinsic Safety & Certification — scored 85/100

> The rating rests on the company's self-reported BIS and PESO certifications for its own manufactured flameproof hydrocarbon transmitters, alongside its role distributing third-party international brands possessing ATEX and IECEx compliance. Based on primary research, the company has not disclosed independent third-party certification documents or direct international safety registrations issued under its own name.

SUPPORTING POINTS

- The company reports that its generic standard 3-wire, 4-20mA hydrocarbon gas sensor system features explosion-proof SS 316L construction certified by BIS and PESO (CCOE) with an Ex db IIC T6 Gb rating.
- The company's product listings include third-party partner products, such as the Gastron GTF-1100U UV/IR flame detector and Simtronics MultiFlame series, which carry ATEX and IECEx certifications.

Environmental Durability & Reliability — scored 96/100

> The assessment rests on the company's detailed technical specifications confirming specific IP-66 ratings and operating temperature tolerances for its proprietary fixed gas detection models. Based on primary research, the company has not disclosed MTBF data or independent field reliability reports.

SUPPORTING POINTS

- The company reports that its PGS-01-IR Infrared Point Type Gas Detector features an SS-316 housing, an operating temperature range of -20º to +70º C, a humidity range of 0 to 99% RH non-condensing, and an ingress protection rating of IP-66.
- The company reports that its Hydrocarbon Gas Detector features an SS-316L sensor material, an LM-6 aluminium enclosure, an ingress protection rating of IP-66, and an operating temperature range of -20º C to +60º C.

Integration & Smart Features — scored 100/100

> The rating is based on confirmed telemetry specifications for the company's proprietary transmitters and multi-channel monitoring systems, which establish clear capabilities for standard RS-485 Modbus serial communication and computer-based software polling. Based on primary research, the company has not disclosed advanced IoT analytics platforms, cloud native remote monitoring architecture, or cellular/wireless smart features for its fixed industrial detection lineup.

SUPPORTING POINTS

- The company reports that its PGS-01 series gas sensors and digital gas transmitters feature an optional RS-485 interface alongside standard 3-wire 4-20mA analog output for integration.
- The company reports that its GMS-4000 gas monitoring system utilizes an intelligent Field Terminal Unit (FTU) and Computer Interface Unit (CIU) architecture running OISD-144 compliant software to digitally transmit analog field sensor signals to a control room computer within 15 seconds.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 97/100

> The assessment rests on the company's self-reported list of major oil & gas, petrochemical, and engineering customers in India alongside general project counts. Based on primary research, the company has not disclosed audited project reference lists, specific unit installation counts, or third-party verified project statuses within the gas and flame detection market.

SUPPORTING POINTS

- The company reports a client list that includes many industry majors such as IOCL, HPCL, IPCL, KRL, BRPL, ONGC, BPCL, Gujarat Gas, BHEL, NRL, OIL, GAIL, Reliance, L&T, Siemens, and ABB.
- The company reports having completed 133 projects and served 95 clients across its product lines, which include gas monitors and electronic sensors.

Revenue & Growth in India — scored 96/100

> The assessment is based on the company's official financial summaries, which reveal stable mid-market revenue figures alongside a slight near-term commercial deceleration. Based on primary research, the company has not disclosed the precise percentage of its total revenue or growth explicitly generated by the gas and flame detection business segment versus its other machinery manufacturing operations.

SUPPORTING POINTS

- Financial records filed with the Ministry of Corporate Affairs reveal that Pentax Engineering Private Limited reported an operating revenue range between INR 10 crore and INR 50 crore for the financial year ending on March 31, 2025.
- Financial statements indicate that the company experienced a net sales revenue decline of 3.52% for the financial year 2025.

Distribution & Service Network — scored 94/100

> We verified that the company maintains its primary operations and central service facility in Mumbai while claiming nationwide customer support coverage. Based on primary research, the company has not disclosed the specific number of domestic sales offices, independent distributors, or localized service centers outside of its Mumbai facility, nor has it published clear metrics regarding service response times.

SUPPORTING POINTS

- The company reports providing nationwide customer support services and handling preventive maintenance and recalibration for customers across India.
- The company's confirmed physical presence, customer service centre, and headquarters are located at a single complex in Mumbai, Maharashtra.

Regulatory Approvals & Tenders — scored 98/100

> The assessment rests on verified procurement data and official safety records demonstrating that the company routinely wins public safety instrumentation tenders and possesses localized statutory compliance through PESO. Based on primary research, the company has not disclosed the complete, multi-year volume or total win-rate metrics for its private and public gas-and-flame tender submissions.

SUPPORTING POINTS

- Pentax Engineering Pvt. Ltd. was awarded a government tender valued at INR 99.12 Lakh in February 2026 for the supply, installation, testing, and commissioning of fire alarm systems at multiple Balol locations.
- The company won a public tender through the Government e-Marketplace (GeM) in August 2026 for multi-gas detectors valued at INR 1.30 Lakh, and has its compliance capabilities verified by an official Petroleum & Explosives Safety Organisation (PESO) approval listing updated in January 2026 as the authorized after-sales service and maintenance representative for Gastron Co. equipment.

Local Manufacturing & Partnerships — scored 84/100

> The rating rests on the company’s documented self-reported local manufacturing presence in Mumbai and its multiple long-standing global technical partnerships for technology sharing. Based on primary research, the company has not disclosed third-party verification details regarding its production volumes or local component sourcing percentages.

SUPPORTING POINTS

- The company reports having in-house manufacturing capabilities for fixed and portable gas detectors, monitors, flame detectors, and monitoring systems/controllers at its facility located in Sakinaka, Mumbai, India.
- The company reports maintaining active technical collaborations with Industrial Scientific Corporation (USA), Oldham SAS (France), Austech Instruments Pty Ltd (Australia), and New Cosmos Electric Co. (Japan) to support their product ranges.

Ambetronics Engineers Pvt. Ltd. Ambetronics ● Leaders X 93 Y 89 Overall 91

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 90/100

> The assessment rests on the company's explicit technical disclosures detailing a versatile suite of advanced sensor integrations across multiple metrics and an established fast response parameter. Based on primary research, the company has not disclosed third-party performance verification data or comparative benchmarks to substantiate these internal accuracy tolerances.

SUPPORTING POINTS

- Ambetronics Engineers Pvt. Ltd. reports that its smart gas detectors (such as the GT-2511-FLP and GT-2500-WP) utilize electrochemical, catalytic bead, pellistor, non-dispersive infrared (NDIR), photoionization detection (PID), and semiconductor sensor technologies.
- The company reports that its fixed industrial detectors achieve a response time of less than 15 seconds, maintaining an accuracy profile of ±2% full scale for electrochemical and catalytic sensors across varying structural metric thresholds like PPM, %LEL, and %V/V.

Product Portfolio Breadth — scored 85/100

> The rating rests on the company's extensive self-reported range of devices, model variants, and gas-sensing capabilities. Based on primary research, the company has not disclosed third-party verification of its total portfolio breadth or independent benchmarking data.

SUPPORTING POINTS

- The company reports offering a comprehensive product portfolio across all major categories, including portable gas detectors, fixed gas detectors/transmitters, multi-spectral UV and infrared flame detectors, gas analyzers, and multi-channel gas monitoring control panels.
- The company reports that its detection systems support multiple sensor technologies—such as electrochemical, catalytic, pellistor, NDIR, PID, and solid-state—to monitor a wide range of target gases, including oxygen, toxic gases, volatile organic compounds, and combustible gases like LPG, CNG, methane, and hydrogen.

Intrinsic Safety & Certification — scored 100/100

> The rating rests on the company's extensive array of localized Indian hazardous-area certifications (PESO, BIS, CIMFR) alongside selective international ATEX, IECEx, and SIL 2 safety ratings for specific fixed gas detector enclosures. Based on primary research, the company has not disclosed independent third-party certification registry links or full certificate validation details directly accessible on its public platform.

SUPPORTING POINTS

- The company reports that its toxic gas detectors and portable models are certified by the Central Institute of Mining and Fuel Research (CIMFR), tested by the Bureau of Indian Standards (BIS), and approved by the Petroleum and Explosives Safety Organization (PESO) for hazardous Zone 1 and Zone 2 environments.
- The company reports that its smart gas detector model GT-2511-FLP features an ATEX and IECEx certified flameproof enclosure compliant with Ex db IIC T6 Gb standards, as well as a SIL 2 functional safety certification under IEC 61508.

Environmental Durability & Reliability — scored 96/100

> The assessment rests on confirmed third-party PESO and CIMFR flameproof compliance certifications alongside the company's self-reported IP ratings and operating temperatures for its gas detection systems. Based on primary research, the company has not disclosed MTBF data or field reliability reports.

SUPPORTING POINTS

- The company reports that its Ambetronics Engineers Pvt. Ltd. GT-4220-FLP gas leak detectors feature an IP68 flameproof enclosure with an operating temperature range of -25°C to +60°C.
- The company reports that its GT-2511-FLP smart gas transmitters feature an IP66/IP68 aluminum or stainless steel (SS-316) housing operating between -15°C and +55°C.

Integration & Smart Features — scored 96/100

> The rating rests on the company's verified provision of standard Modbus and RS-485 protocols across its gas detector lineup, alongside its standalone cellular and gateway hardware options for cloud and IoT telemetry. Based on primary research, the company has not disclosed third-party verified native integrations with specific large-scale PLC or SCADA software ecosystems, nor advanced built-in edge data analytics capabilities.

SUPPORTING POINTS

- The company reports that its wireless battery-powered gas detector (WGD-100-FLP) features 4G GSM connectivity with dual SIM slots, transmitting data to a wireless control unit over Modbus RTU/Modbus TCP-IP and utilizing MQTT protocol for cloud connectivity.
- The company reports that its smart gas detectors (such as the GT-2511-FLP) include standard 4-20mA signal outputs alongside an optional isolated RS-485 communication port running Modbus RTU protocol for control system integration.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 99/100

> The rating rests on the company's self-reported list of major customer accounts within industrial and petrochemical segments in India, including prominent paint and energy corporations. Based on primary research, the company has not disclosed the precise count of its active installations or unit volume in the gas and flame detection market.

SUPPORTING POINTS

- The company reports serving major accounts including PPG Industries, Asian Paints, HPCL, and IOCL.

Revenue & Growth in India — scored 69/100

> Based on primary research, the company has not disclosed granular segment-wise sales revenue, year-over-year market growth, or market share data for its gas and flame detection systems.

SUPPORTING POINTS

- Not available

Distribution & Service Network — scored 100/100

> The rating rests on the company's established, multi-location domestic branch network across major industrial hubs and its self-reported 24/7 technical and on-site after-sales support capabilities. Based on primary research, the company has not disclosed audited service turnaround metrics, quantified distributor counts, or third-party verified response times.

SUPPORTING POINTS

- The company reports its registered office and factory are located in Mumbai, Maharashtra, along with branch offices in Ahmedabad, Bangalore, Bharuch, Chennai, Delhi, Hyderabad, Kolkata, Pune, and Vapi.
- The company reports providing PAN India after-sales support, 24/7 customer support, and on-site services including calibration, installation, commissioning, and preventive maintenance across multiple customer sites.

Regulatory Approvals & Tenders — scored 98/100

> The evaluation is based on confirmed public tender wins with state-owned enterprises and local safety certifications for the company's hardware line. Based on primary research, the company has not disclosed the complete volume or total contract value of its private market tender engagements.

SUPPORTING POINTS

- The company reports that its gas detection products, including models like the GT-4220-FLP and GT-2511-FLP, are certified by the Central Institute of Mining and Fuel Research (CIMFR) and hold Petroleum and Explosives Safety Organization (PESO) flameproof enclosure approvals.
- Government e-Marketplace (GeM) records show that the company was awarded a ₹4.95 Lakh public tender by the Ministry of Petroleum and Natural Gas for the procurement of 44 handheld CNG detectors for Indian Oil Corporation Limited.

Local Manufacturing & Partnerships — scored 80/100

> The rating rests on the company's verified operational presence and self-reported local manufacturing capabilities for gas and flame detection systems in Mumbai. Based on primary research, the company has not disclosed audited details regarding external technology transfer agreements or its local component sourcing ratios.

SUPPORTING POINTS

- The company reports operating an integrated manufacturing and designing facility at its registered office in Mumbai, Maharashtra, which features an installed production capacity of 1,000 units per month utilizing modern calibrated equipment.
- The company reports that its production unit is manned by experienced engineers and technicians who assemble gas and flame detection instruments utilizing component materials like printed circuit boards and integrated circuits.

Victory Gas Alarm Company Victory Challengers X 82 Y 90 Overall 86

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 81/100

> We verified that the company's technical capabilities rest on self-reported documentation for specific handheld and domestic safety detectors featuring basic catalytic and electrochemical sensor configurations. Based on primary research, the company has not disclosed comprehensive, third-party certified testing or advanced technological parameters such as infrared technology across its industrial portfolio.

SUPPORTING POINTS

- The company reports its V 9000 Combustible Gas Detector uses catalytic combustion and electrochemical sensor types to measure concentrations of toxic, oxygen, and combustible gases (PNG/LPG/CNG) over a range of 0 to 100% LEL or parts-per-million (ppm), achieving a response time of T90 < 10 seconds.
- The company reports its V-201 domestic gas detector utilizes a catalytic combustion sensor targeting combustible gases with an alarm range of 0.1 to 0.3% volume and an alarm level set at 10% LEL.

Product Portfolio Breadth — scored 76/100

> The rating reflects a foundational domestic and light-industrial product range spanning portable hand-held devices, fixed sensor units, and multi-channel monitoring panels across common combustible and toxic targets. Based on primary research, the company has not disclosed flame detection systems or advanced industrial analytical automation lines within its public catalog.

SUPPORTING POINTS

- The company reports offering portable gas detectors (such as the V-700 and pen-type variants), fixed units (like the V205D), and centralized control hardware including the multi-channel V-8000 digital controller and KB6000 III multi-point gas detection panel.
- Product commercial documentation notes that these systems track combustible and toxic compounds including LPG, PNG, CNG, carbon monoxide, methane, ammonia, and hydrogen sulfide.

Intrinsic Safety & Certification — scored 88/100

> We verified that the company maintains core regional capability with certified Petroleum and Explosives Safety Organisation (PESO) approvals and Central Mining Research Institute (CMRI) testing for hazardous areas. Based on primary research, the company has not disclosed comprehensive international certifications such as ATEX or IECEx certificates on its standard product lines.

SUPPORTING POINTS

- Victory Gas Alarm Company reports that its GT-18 combustible gas detector, V-4000 single gas detector, and V-205DEx combustible gas detector are PESO approved.
- The company has successfully developed Intrinsically Safe Gas Detectors for Combustible Gas and Vapour that were tested by the Central Mining Research Institute (CMRI).

Environmental Durability & Reliability — scored 93/100

> The rating rests on the company's baseline specification disclosures for select products, such as IP66 protection and wide operating temperature ranges on its handheld units. Based on primary research, the company has not disclosed comprehensive, third-party verified environmental endurance test data, field reliability records, or mean time between failures (MTBF) statistics.

SUPPORTING POINTS

- The company reports an ingress protection rating of IP66 and an operating temperature range of -30°C to +50°C for its V-700 hand-held gas detector.
- The company reports an operating temperature range of -20°C to +50°C for its V-2000 digital gas alarm panel, and uses shock-proof, weather-resistant enclosures for its 205D fixed model.

Integration & Smart Features — scored 73/100

> The rating reflects the company's demonstrated production of multi-protocol gas alarm panels, fixed digital gas transmitters, and wired controllers equipped with programmable alert stages and relay logic automation capabilities. Based on primary research, the company has not disclosed explicit software platforms for advanced data analytics or native IoT remote monitoring integrations.

SUPPORTING POINTS

- Victory Gas Alarm Company reports that its VAC6000I Gas Alarm Panel incorporates multi-protocol configurations and advanced functionalities.
- Victory Gas Alarm Company manufactures fixed digital gas transmitters and multi-channel gas control panels designed to process signals and trigger local audio-visual alerts or relay automation commands.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 97/100

> The assessment rests on the company's self-reported historical project counts and high-level sector affiliations alongside tracked government bidding activity. Based on primary research, the company has not disclosed a verified list of active industrial installations, exact customer project references, or specific installed unit counts in India.

SUPPORTING POINTS

- Victory Gas Alarm Company reports completing over 5,000 projects and serving more than 2,000 clients globally.
- The company reports its gas detectors and systems are utilized by the Indian Railways, defense organizations, and petrochemical industries, and it records public tender activity with buyers such as Indianoil and Rashtriya Chemicals and Fertilizers Ltd.

Revenue & Growth in India — scored 85/100

> The rating rests on public tender data showing limited domestic government contract revenue, which represents only a subset of total operations. Based on primary research, the company has not disclosed its comprehensive annual sales revenue, financial reports, or year-over-year growth rates for the Indian market.

SUPPORTING POINTS

- The company won 8 of 99 government tenders held in public opened-bid records, achieving a recorded won contract value of ₹15.4 Lakh.

Distribution & Service Network — scored 94/100

> The rating rests on the company's self-reported Pan-India presence spanning key cities like Mumbai and Delhi and regional authorized distribution partnerships. Based on primary research, the company has not disclosed the exact total number of distribution points, service centers, or specific service response timelines.

SUPPORTING POINTS

- The company reports a "PAN INDIA SERVICE NETWORK" including locations in Mumbai and Delhi, along with an authorized distributor for Maharashtra and Goa.

Regulatory Approvals & Tenders — scored 95/100

> The rating rests on verified government tender wins for hazardous gas detection equipment by the company from Indian public sector enterprises and railways. Based on primary research, the company has not disclosed comprehensive documentation of its standalone PESO approval certifications or international product compliance standards.

SUPPORTING POINTS

- The company won a tender from North Eastern Railway on June 30, 2026, for the supply and installation of a fire explosion meter (explosive gas detector) worth ₹30,090.
- The company won multiple public tenders from The Fertilisers and Chemicals Travancore Ltd for the procurement of portable gas detectors (valued at ₹6.7 Lakhs) and multigas detectors (valued at ₹5.3 Lakhs).

Local Manufacturing & Partnerships — scored 77/100

> The rating rests on the company's self-reported long-standing status as a domestic manufacturer supplying public sector entities like the Indian Railways. Based on primary research, the company has not disclosed independent verification of its manufacturing infrastructure capacity or active strategic corporate partnerships.

SUPPORTING POINTS

- The company reports operating as a manufacturer of gas detectors, gas alarm panels, and automatic fire detection systems in India since 1988.
- The company claims to be the first in India to obtain a patent for its gas detectors, which are supplied to organizations like the Indian Railways and defense agencies.

Addict Automation (India) Private Limited GAS JANITOR ● Challengers X 78 Y 89 Overall 84

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 75/100

> The evaluation rests on the company's verified self-reported capability to manufacture industrial fixed and portable single-gas detectors under its Gas Janitor brand that cover basic semiconductor configurations for targets such as LPG, methane, and toxic gases up to 1000 ppm. Based on primary research, the company has not disclosed third-party performance verifications or data regarding advanced infrared (IR) or catalytic sensor varieties.

SUPPORTING POINTS

- The company reports technical specifications for its Model GJ-0417 Flameproof Methane Gas Detector, which utilizes a semiconductor sensor operating via diffusion to detect methane with a response time of 15 seconds and an alarm threshold of 10% LEL, according to its product specifications listed on IndiaMART.
- The company reports a Toxic Gas Detection System featuring a backlit digital LCD display with a measuring range spanning 0–1000 ppm to monitor hazardous industrial and commercial gases, according to product documentation sourced via Justdial.

Product Portfolio Breadth — scored 76/100

> The rating rests on the company's established self-reported line of fixed gas leak detectors and microprocessor-based alarm control panels. Based on primary research, the company has not disclosed any optical or specialized flame detectors or portable gas detection models in its current portfolio.

SUPPORTING POINTS

- The company reports offering permanently fixed weatherproof and flameproof industrial and commercial gas leak detectors, such as the GJ-0517, GJ-0317, and the PESO-approved GJ-0417 and GJ-0617 models, which support gases including LPG, PNG, Ammonia, and Hydrogen, as published on its official Gas Janitor Brand Portal.
- The company reports supplying microprocessor-based conventional type gas leakage control panels with 2 zones and 16x2 LCD line displays, as detailed in its product technical specifications listed on Justdial.

Intrinsic Safety & Certification — scored 79/100

> The assessment is based on the company's self-reported alignment with standard Indian hazardous area compliance protocols across its localized product catalog. Based on primary research, the company has not disclosed international explosive environment certifications such as ATEX or IECEx.

SUPPORTING POINTS

- The company's digital-type flameproof gas detector, Model GJ-0617 under its Gas Janitor brand, holds domestic safety compliance via PESO, CMRI, and BIS certifications alongside an IP66 enclosure rating, according to product documentation on IndiaMART.
- The company's permanently fixed industrial gas detector, Model GJ-0417 designed for LPG, PNG, ammonia, and hydrogen gases, carries a PESO/CCOE approved IP66 flameproof rating, as reported on the official Gas Janitor platform.

Environmental Durability & Reliability — scored 96/100

> We verified that the company's hardware provides certified IP66 ingress protection and established operational ranges for temperature and humidity. Based on primary research, the company has not disclosed MTBF data or field reliability reports to confirm long-term operational performance or false alarm metrics.

SUPPORTING POINTS

- The company's flameproof gas detectors under the Brand Gas Janitor feature an IP66 enclosure class protection, an operational environmental temperature range of 0 to 55°C, and an environmental humidity tolerance of up to 95% non-condensing.
- Product documentation lists technical safety and environmental compliance certifications from independent bodies including CMRI, BIS, and PESO (CCOE).

Integration & Smart Features — scored 65/100

> The evaluation rests on the company's baseline product documentation indicating basic wired compatibility with external control units and localized systems. Based on primary research, the company has not disclosed advanced integration and smart features such as wireless connectivity, remote monitoring, data analytics software, or standard PLC and SCADA communication protocols.

SUPPORTING POINTS

- The company reports that its fixed gas leakage detectors, including the GJ0617 hydrogen and GJ0417 ammonia models, connect to a control panel via four-core cables and are compatible with Building Management Systems (BMS) and fire panels with customization.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 95/100

> We verified that the company is listed as an advisory contractor for a major city gas distribution utility. Based on primary research, the company has not disclosed its total installed unit counts or a comprehensive list of its active industrial accounts.

SUPPORTING POINTS

- Indraprastha Gas Limited included Addict Automation (India) Private Limited on its advisory list of contractors for executing internal works as an approved supplier of gas detectors and solenoid valves, published on August 29, 2026.

Revenue & Growth in India — scored 91/100

> The rating rests on the company's confirmed small-scale revenue band under official corporate records. Based on primary research, the company has not disclosed its exact sales figures, year-over-year revenue growth rates, or market share metrics in the Indian gas and flame detection market.

SUPPORTING POINTS

- The company reports generating less than ₹10 Cr in revenue for the financial year ending on March 31, 2025.

Distribution & Service Network — scored 92/100

> The evaluation rests on the company's disclosure of two corporate office footprints and self-reported nationwide installation capabilities. Based on primary research, the company has not disclosed information regarding an authorized distributor network, dedicated service center counts, or certified service response times.

SUPPORTING POINTS

- The company reports its setup across two distinct facilities in India, consisting of a Delhi office situated in the Kirti Nagar Industrial Area, New Delhi, and an Agra office located on Shani Dev Marg, Runakta, Agra.
- The company reports providing its gas leak detectors installation services with a service location coverage designated as PAN India.

Regulatory Approvals & Tenders — scored 90/100

> The assessment rests on the company's self-reported technical compliance and regulatory certification for its product line under the Gas Janitor brand. Based on primary research, the company has not disclosed any verified public or private tender wins or formal contract awards in the gas and flame detection market.

SUPPORTING POINTS

- The company reports offering fixed single gas digital flameproof gas detectors under its brand Gas Janitor that comply with PESO (CCOE), CMRI, and BIS certifications.

Local Manufacturing & Partnerships — scored 77/100

> We verified that the entity operates local production and assembly facilities in India for its gas detection brand. Based on primary research, the company has not disclosed strategic joint ventures, technology transfers, or local component sourcing depths.

SUPPORTING POINTS

- The company reports operating as a manufacturer of gas leak detectors and control panels under its brand name Gas Janitor, with facilities located in the Kirti Nagar and Rama Road industrial areas of New Delhi.
- The company's industrial gas monitoring products have received certifications from Indian regulatory bodies, including the Petroleum and Explosives Safety Organisation, Central Mining Research Institute, and Bureau of Indian Standards.

Joseph Leslie Dynamiks Manufacturing Pvt. Ltd. Joseph Leslie Challengers X 76 Y 91 Overall 84

X — PRODUCT CAPABILITY

Intrinsic Safety & Certification — scored 65/100

> We verified that the company mentions offering testing aligned with PESO and IS standards on its service portfolio. Based on primary research, the company has not disclosed specific intrinsic safety certificates or ratings (such as ATEX or IECEx numbers) for its proprietary gas and flame detection hardware line.

SUPPORTING POINTS

- The company reports providing certified testing and inspection in accordance with PESO and IS standards through its integrated services, according to the Joseph Leslie Dynamiks Services Page.

Environmental Durability & Reliability — scored 92/100

> The assessment rests on self-reported, broad compliance certifications (ATEX/IECEx temperature classes T4/T5) for the company's fixed gas systems. Based on primary research, the company has not disclosed specific environmental ingress protection (IP) ratings, quantitative mean time between failures (MTBF) statistics, or documented field reliability case studies for its dedicated gas and flame detection devices.

SUPPORTING POINTS

- The company reports that its Fixed Gas Detector line is designed for continuous monitoring in explosive industrial atmospheres with certified variants supporting Zone 1 and Zone 2, Gas Groups IIA, IIB, and IIC, and a temperature class up to T4 / T5, according to the official Joseph Leslie Dynamiks Fixed Gas Detection Catalog.

Integration & Smart Features — scored 65/100

> We verified that the company's public documentation details basic fixed and portable gas monitoring with real-time visual and audible alarm configurations. Based on primary research, the company has not disclosed specific communication protocols, software platform compatibility, or smart integration capabilities with PLC, SCADA, or IoT networks for its standalone gas and flame detection hardware.

SUPPORTING POINTS

- The company reports that its fixed gas detection solutions, including the JLD IR / TOX fixed detection system, offer real-time gas concentration readings and visual status indication for industrial and hazardous areas, as presented on its IndiaMART Brand Profile and official product descriptions.

Sensor Technology & Detection Range — scored 81/100

> We verified that the company manufactures fixed and portable systems incorporating common sensor types like electrochemical, catalytic, PID, and IR. Based on primary research, the company has not disclosed specific technical metrics including detection limits, exact measurement ranges (LEL, ppm, %vol), or numeric response times.

SUPPORTING POINTS

- The company reports that its fixed gas detection systems feature stable and repeatable sensor performance, suitable for Zone 1 and Zone 2 gas groups (IIA, IIB, IIC) with intrinsically safe (Ex i) and flameproof (Ex d) variants.
- The company reports providing on-site and laboratory calibration services specifically for infrared (IR), electrochemical, catalytic bead, and photoionization detector (PID) sensors.

Product Portfolio Breadth — scored 76/100

> The assessment is based on the company's verified product categories covering portable detectors, fixed systems, flame detection, and tubes. Based on primary research, the company has not disclosed the precise number of unique models, specific target gas types covered, or control panel specifications across these product categories.

SUPPORTING POINTS

- The company reports offering a product lineup that includes portable gas detection monitors, fixed gas detection systems, flame detection devices, emission analysers, and Gastec detector tubes.
- The company reports supplying equipment both from its own manufacturing facility in Mumbai and through international business partners, such as GasTech Australia.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 87/100

> We verified that the company has a long operational history serving high-risk sectors such as oil and gas, petrochemicals, pharmaceuticals, mining, utilities, and defence. Based on primary research, the company has not disclosed customer names, project references, or installed unit counts.

SUPPORTING POINTS

- The company reports providing its entire range of gas detection equipment through its own state-of-the-art facility in Mumbai as well as from its international partner GasTech Australia.

Revenue & Growth in India — scored 98/100

> The evaluation rests on independent financial highlights confirming a year-over-year revenue increase of over 22% for the company in 2025. Based on primary research, the company has not disclosed its exact absolute revenue figures or segmented market share metrics specifically for the gas and flame detection market.

SUPPORTING POINTS

- Joseph Leslie Dynamiks Manufacturing Private Limited reported a net sales revenue increase of 22.28% in 2025.

Distribution & Service Network — scored 96/100

> The assessment rests on the company's confirmed presence across major industrial hubs in India through a corporate office, a manufacturing factory, and five distinct regional branch offices. Based on primary research, the company has not disclosed the exact count or operational locations of its distribution network partners or its specific service response times.

SUPPORTING POINTS

- The company reports being headquartered in Mumbai with branch offices in Delhi, Chennai, Baroda, Chandigarh, and Kolkata.
- The company reports employing a team of over 120 trained professionals and maintaining an extensive nationwide distributor network to support service and installation across India.

Regulatory Approvals & Tenders — scored 91/100

> We verified that the company has actively participated in and won multiple public procurement tenders in India for industrial safety systems, including a contract from GAIL India Limited awarded on 26-03-2026 for breathing apparatus sets. Based on primary research, the company has not disclosed specific PESO approvals or separate tender wins explicitly for its gas and flame detection devices.

SUPPORTING POINTS

- The company reports obtaining Bureau of Indian Standards (BIS) approval for its breathing apparatus sets in 2004 and for its full face masks in 2008.

Local Manufacturing & Partnerships — scored 84/100

> Assessment rests on the company's verified operations at its local assembly and manufacturing site in Maharashtra alongside disclosed global technical distribution partnerships. Based on primary research, the company has not disclosed specific third-party verification details of local production volume shares or complete technology-transfer contracts for its gas and flame detection systems.

SUPPORTING POINTS

- The company reports operating a state-of-the-art manufacturing facility in North Mumbai (Vasai East, Palghar) for the production of gas detection and respiratory protection solutions.
- The company reports strategic international business partnerships for its product portfolio with global entities including Gastech, Cressi Sub, ISG Infrasys, Nardi, and O-Two.

Trace Automation Pvt. Ltd. Trace Automation ● Challengers X 85 Y 81 Overall 83

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 81/100

> The assessment rests on the company's self-reported technical specifications for its specific gas monitoring systems and flame detectors, detailing response times, accuracy levels, and sensor technologies. Based on primary research, the company has not disclosed independent third-party performance verification data or comprehensive detection limits across all deployed electrochemical and catalytic sensor models.

SUPPORTING POINTS

- The company reports technical specifications for its Model RS24IG-FLP Hazardous Gas Monitoring System, claiming a measurement range of 0–1,000 ppm, accuracy of 2% full scale, zero drift of 0.2% full scale, and a response time of less than 5 seconds using a PID sensor type.
- The company reports its "Trace" brand UV IR Flame Detector System features a triple IR spectral range, a response time of under 3 seconds, a detection range of 30 meters, and utilizes a UV sensor.

Product Portfolio Breadth — scored 80/100

> The rating rests on the company's self-reported range of portable and fixed gas detectors, centralized multi-channel monitoring panels, and flame detector systems covering diverse toxic and combustible gases. Based on primary research, the company has not disclosed independent third-party market validation or quantified model counts across all stated applications.

SUPPORTING POINTS

- The company reports a product line that includes portable single and multi-gas detectors, fixed gas detectors (such as the RS24IG-FLP model), multi-channel centralized gas monitoring panels, and UV/IR flame detector systems.
- The company reports that its hazardous gas monitoring systems cover a wide range of gases, including ammonia (NH_3), hydrogen sulfide (H_2S), chlorine (Cl_2), carbon monoxide (CO), sulfur dioxide (SO_2), phosphine (PH_3), nitrogen dioxide (NO_2), chlorine dioxide (ClO_2), hydrogen cyanide (HCN), hydrochloric acid (HCl), oxygen, and various combustible gases.

Intrinsic Safety & Certification — scored 81/100

> The rating rests on the company's self-reported explosion-proof and hazardous area certifications (PESO, ATEX, IECEx, CMRI, and ISI) for its fixed gas leak detection models. Based on primary research, the company has not disclosed independent third-party certificate copies or official registry verifications to validate these compliance claims.

SUPPORTING POINTS

- The company reports that its fixed gas detector model RS24IG-FLP features ATEX, IECEx, and PESO certifications.
- The company reports that its fixed gas detector model RS24IGPRO-FLP-LED features explosion-proof certifications including PESO, CMRI, and ISI.

Environmental Durability & Reliability — scored 95/100

> The rating rests on the company's disclosed environmental specifications for select equipment, including operating temperatures, humidity boundaries, and standard IP enclosure ratings. Based on primary research, the company has not disclosed mean time between failures data or comprehensive field reliability reports detailing long-term industrial operations and false alarm frequencies.

SUPPORTING POINTS

- The company reports its residential gas leak detector operates within a temperature range of -10°C to +55°C and an operating humidity range of 10% to 90% RH (non-coagulating).
- Product listings show the company's SS316L Alphanumeric LCD Hydrocarbon gas detector features an IP55 rating for its sensor housing and an IP65 or IP66 rating for its flameproof enclosure.

Integration & Smart Features — scored 87/100

> The rating rests on confirmed product specifications showing that the company's gas monitoring panels support standard industrial interfaces like Modbus RS-485, 4-20mA signals, and integration compatibility with external PLC, SCADA, and BMS networks. Based on primary research, the company has not disclosed advanced smart features such as proprietary software platforms, cloud-based data analytics, or wireless IoT connectivity for its gas and flame detection line.

SUPPORTING POINTS

- The company reports that its Multi Channel Gas Monitoring Panel can be integrated with external control equipment including PLC, SCADA, BMS, and Emergency Shutdown Systems.
- The company reports that its multi-channel gas monitoring system supports 4-20mA, Relay, and Modbus RS-485 communication protocols for interfacing with current infrastructure.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 65/100

> Based on primary research, the company has not disclosed its customer list, project references, or installed unit counts.

SUPPORTING POINTS

- Not available

Revenue & Growth in India — scored 83/100

> Based on primary research, the company has not disclosed segment-specific sales revenue, year-over-year growth metrics, or market share data for the gas and flame detection market in India.

SUPPORTING POINTS

- The company reported an annual turnover range of ₹1.5 Crore to ₹5 Crore on its trade profile directory listings.

Distribution & Service Network — scored 86/100

> The assessment rests on the company's verified primary manufacturing operations in Pune alongside regional branch presence across two additional hubs in India. Based on primary research, the company has not disclosed the precise count of independent distributor networks, dedicated service centers, or explicit service turnaround times.

SUPPORTING POINTS

- The company reports its main office and infrastructure facility are located in Guruwar Peth, Pune, Maharashtra.
- Commercial business listings document supplementary physical operational footprints in Nagpur, Maharashtra, and Ambattur, Chennai.

Regulatory Approvals & Tenders — scored 91/100

> The rating rests on a 2026 government tender win we verified for plant systems maintenance and fixed H2S sensor installation alongside company reports of holding domestic PESO certifications. Based on primary research, the company has not disclosed a comprehensive portfolio of private sector contract wins or direct copies of its official regulatory approval certificates.

SUPPORTING POINTS

- Trace Automation Private Limited was awarded a ₹24 Lakh public tender in September 2026 for the repair, maintenance, and installation of plant systems, which included installing fixed hydrogen sulfide (H2S) sensors in the Andaman and Nicobar region.
- The company reports that its industrial gas detection transmitters and monitoring systems, including the RS24IGDH2-FLP-LED model, carry Petroleum and Explosives Safety Organisation (PESO) and CMRI certifications for hazardous environments.

Local Manufacturing & Partnerships — scored 80/100

> We verified that Trace Automation Pvt. Ltd. maintains its own local manufacturing and assembly presence in Pune alongside commercial ties to supply multinational brands like Honeywell. Based on primary research, the company has not disclosed audited local sourcing volumes or formalized joint venture agreements.

SUPPORTING POINTS

- The company reports operating an infrastructure facility in Pune, Maharashtra, that includes dedicated production, quality testing, and research and development departments for gas leak detectors and monitoring systems.
- The company reports a trading and supply association with multinationals and Original Equipment Manufacturers (OEMs) such as Honeywell to deliver gas safety technologies to the Indian market.

R. J. Electricals Private Limited Tinselpro ● Challengers X 76 Y 84 Overall 80

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 65/100

> The rating rests on the company's verified manufacture of basic portable and wall-mounted gas detectors utilizing semiconductor and electrochemical sensors with reported response times. Based on primary research, the company has not disclosed comprehensive detection limits, full response accuracy metrics across its diverse industrial gas variants, or any implementation of advanced infrared (IR) or catalytic sensor technologies.

SUPPORTING POINTS

- The company reports that its Industrial Flameproof Methane (CH4) Gas Leak Detector (Model TP-3051-D) utilises a semiconductor sensor type and delivers a response time of 3 to 5 seconds.
- The company reports manufacturing a handheld portable gas leak detector that integrates an electrochemical sensor, alongside products featuring a digital microprocessor-based concentration display measured in parts per million (ppm).

Product Portfolio Breadth — scored 76/100

> The rating rests on the company's verified range of portable gas detectors, fixed systems, and control panels covering a broad spectrum of toxic and combustible gases. Based on primary research, the company has not disclosed any proprietary models or dedicated standalone optical flame detectors within its safety equipment portfolio.

SUPPORTING POINTS

- The company reports a gas detection portfolio under its "Tinsel" and "Tinselpro" brands that includes portable, wall-hung domestic, weatherproof commercial, and flameproof industrial gas leak detectors.
- The company reports that its gas leak detectors cover various gas types including LPG, PNG, LNG, CNG, Methane (CH4), Ammonia (NH3), Carbon Monoxide (CO), Carbon Dioxide (CO2), Oxygen (O2), Hydrogen, LEL, and CFCs, alongside offering industrial control panels (e.g., model TPCP04Z).

Intrinsic Safety & Certification — scored 77/100

> The assessment is based on the company's self-reported Indian hazard approvals (PESO, CCOE, and CMRI) for its industrial flameproof gas detector housings. Based on primary research, the company has not disclosed third-party verification certificates or international compliance listings such as ATEX or IECEx.

SUPPORTING POINTS

- The company reports that its Industrial Flameproof LPG/PNG Gas Leak Detector (Model TP-3051 - ND) features a flameproof housing that is CCOE, PESO, and CMRI approved.
- The company reports that it holds an ISO 9001:2015 standard certification for its manufacturing operations.

Environmental Durability & Reliability — scored 93/100

> Our assessment rests on the company's verified IP65 ingress protection rating for industrial units and self-reported temperature and humidity parameters for commercial lines. Based on primary research, the company has not disclosed MTBF data or field reliability reports.

SUPPORTING POINTS

- The company's TINSELPRO industrial flameproof gas leak detectors feature an IP65 ingress protection rating.
- The company reports that its commercial weatherproof gas leak detectors operate within a temperature range of 0 to 55°C and handle humidity levels below 95%.

Integration & Smart Features — scored 69/100

> The evaluation rests on the company's verified capability to manufacture standalone microprocessor-based detectors that provide output relay actions for basic automated alarming. Based on primary research, the company has not disclosed compatibility with advanced industrial communication protocols, software platforms, or integrations with external PLC, SCADA, or IoT networks.

SUPPORTING POINTS

- The company reports that its gas leak detectors (marketed under the brand "Tinsel" / "Tinsel Pro") are digital, microprocessor-based systems that feature a digital PPM concentration display, built-in automatic alarming, and an integrated output relay signal.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 100/100

> The rating rests on the company's extensive self-disclosed customer list spanning major public infrastructure bodies, hospitality brands, and heavy manufacturers. Based on primary research, the company has not disclosed audited installation metrics or third-party verified deployment counts across these accounts.

SUPPORTING POINTS

- The company reports supplying its safety and gas leak alarming systems to prominent entities including BHEL, ISRO, and Indian Railways.
- The company reports serving customers across diverse industrial sectors, including hospitality clients such as Taj, ITC, Hyatt, DoubleTree by Hilton, Shangri-La, Holiday Inn, Royal Plaza, and Crowne Plaza, alongside industrial and manufacturing accounts like Yamaha Motors, Bajaj Ecotech, Mechwell Industries, Navodaya Minerals, Parle, Asco, Honda, Orient Fans, Maruti, and Adani.

Revenue & Growth in India — scored 71/100

> The rating rests on public business listing estimates showing an approximate annual turnover between ₹40 Lakh and ₹1.5 Crore for the company. Based on primary research, the company has not disclosed audited financial reports, historical year-over-year revenue growth rates, or breakdown data for its gas and flame detection segments in the Indian market.

SUPPORTING POINTS

- R. J. Electricals Private Limited has an estimated annual turnover ranging from ₹40 Lakh to ₹1.5 Crore.

Distribution & Service Network — scored 82/100

> The assessment rests on the company's established footprint in New Delhi and generalized self-reported statements regarding a nationwide service reach. Based on primary research, the company has not disclosed the exact count of its distributors, regional branch offices, authorized service centers, or certified response times.

SUPPORTING POINTS

- Third-party trade portals indicate that R. J. Electricals Private Limited operates from a primary manufacturing and registered office location in New Delhi, India.
- The company reports a pan-India customer base and serves multiple domestic regions through its sales network.

Regulatory Approvals & Tenders — scored 89/100

> The rating rests on the company's self-reported regulatory approvals from PESO and CMRI, alongside a stated history of providing gas detection systems to prominent public enterprises. Based on primary research, the company has not disclosed independent verification certificates or specific government tender award documentation verifying concrete contract metrics.

SUPPORTING POINTS

- The company reports that its TINSELPRO series of fixed single gas monitors and flameproof housings possess CCOE, PESO, and CMRI testing approvals.
- The company reports supplying industrial safety equipment to public sector entities including BHEL, ISRO, and Indian Railways.

Local Manufacturing & Partnerships — scored 80/100

> The rating rests on the company's established domestic production footprint in New Delhi for its proprietary gas leak detectors, utilizing high-end manufacturing techniques to cater to industrial safety clients across India. Based on primary research, the company has not disclosed third-party verification of its production capacity, localized supply chain distributions, or formal technology transfer partnerships.

SUPPORTING POINTS

- The company reports operating advanced manufacturing facilities at Bawana Industrial Area in New Delhi, India, to produce its "TINSEL" and "TINSELPRO" branded digital and flameproof gas leak detection systems.

Pollution Protection Systems Mumbai Pvt. Ltd. PPS ● Trailblazers X 91 Y 86 Overall 88

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 81/100

> The rating rests on the company's demonstrated ability to manufacture and supply both fixed and portable detection systems leveraging standard catalytic beads and electrochemical cells across typical LEL and ppm ranges. Based on primary research, the company has not disclosed comprehensive performance verifications or response time details across its entire internal sensor range, often relying on distributed partner technologies for its high-end open-path line items.

SUPPORTING POINTS

- The company reports that its gaZsens Ex-I and Ex-WP combustible gas sensors utilize poison-resistant catalytic beads to measure ranges of 0–100% LEL, offering a typical sensor lifetime of five years.
- The company reports that its gaZsens Txi toxic gas detector utilizes electrochemical cell technology to measure gases such as H2S, NH3, and Cl2 across selectable ranges of 0–20, 0–50, or 0–00 ppm.

Product Portfolio Breadth — scored 85/100

> We verified that the company manufactures and distributes an extensive selection of portable detectors, fixed gas monitoring systems, online analyzers, flame detectors, and control cards covering multiple industrial applications. Based on primary research, the company has not disclosed the precise total number of active models within each line or a standardized breakdown of all supported target gas types.

SUPPORTING POINTS

- The company reports a comprehensive line of gas and flame detection products including portable gas detectors (gaZguard), online gas analyzers (gaZpro), continuous gas monitoring systems (gaZalarm with gaZsens), stack monitoring systems, personal gas samplers, and specialized instruments for natural gas leak surveys.
- The company reports offering fixed ATEX/IECEx and SIL 2 approved gas detectors, flame monitoring systems, control panels (such as the gaZScan scanner-based monitoring system and gaZalarm control cards), and distributed trading brands like E2S warning signals and Signaline heat sensing cables.

Intrinsic Safety & Certification — scored 100/100

> We verified that the company's rating rests on documented domestic and international explosion-proof certifications (PESO, CIMFR, ATEX, IECEx) for its key fixed gas detection series. Based on primary research, the company has not disclosed the specific certificate documentation numbers or direct download links for individual product compliance files on its public portal.

SUPPORTING POINTS

- Pollution Protection Systems Mumbai Pvt. Ltd. manufactures gas detection lines including the "gaZsens Series" and "IXD 2100 Series" that carry dual domestic and international certifications, specifically CIMFR and PESO for domestic configurations alongside ATEX, IECEx, and CE approvals for explosive environments.
- The company reports that its specialized components, such as the gaZsens Txi toxic gas detector and gaZsens Ex-I combustible gas sensor, hold flameproof Ex-d approvals (IS 2148:2004 / IEC 60079-1) and weatherproof IP55/IP66 ratings certified by CIMFR, Dhanbad.

Environmental Durability & Reliability — scored 98/100

> The rating rests on the company's clear technical specifications for enclosure protection, hazardous area certifications, and thermal performance ranges. Based on primary research, the company has not disclosed MTBF data or independent field reliability reports.

SUPPORTING POINTS

- The company reports that its gaZsens series and IXD 2100 series gas detectors are built with a rugged IP66 dustproof and waterproof aluminum die-cast housing and an SS316 grade stainless steel sensor housing, operating within a temperature range of -40°C to +60°C.
- The company reports that its gas detector systems have received certified safety performance credentials including SIL-2 compliance under IEC-61508 along with ATEX, IECEx, CIMFR, and PESO hazard approvals.

Integration & Smart Features — scored 92/100

> We verified that the company's capabilities center on standard wired automation interfaces, including 4-20mA analog signal outputs and optional RS485 Modbus RTU protocol for control system integration. Based on primary research, the company has not disclosed proprietary cloud IoT monitoring platforms, software analytics suites, or native wireless connectivity modules.

SUPPORTING POINTS

- The company reports offering PLC and HMI-based fixed gas monitoring panels and systems.
- The company reports that its gaZalarm control cards provide 4-20mA repeater signal outputs for direct interface with industrial PLC or DCS architectures, while its gazScan microcontroller-based system supports optional RS485 Modbus RTU communication protocols.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 97/100

> We verified that the company's evaluation rests on self-reported operational capabilities across key manufacturing, refinery, and gas distribution industries, highlighted by its claim of serving over 100 global LPG facilities. Based on primary research, the company has not disclosed specific verified customer names, contract dates, project statuses, or precise installed unit counts within India.

SUPPORTING POINTS

- The company reports having equipped more than 100 LPG facilities around the world with customized gas detection systems.
- The company reports providing flame and gas detection systems across target Indian industrial sectors, including oil and gas, city gas distribution, chemicals and pharmaceuticals, steel, and fertilizer.

Revenue & Growth in India — scored 69/100

> Based on primary research, the company has not disclosed its exact sales revenues, year-over-year market growth rates, or market share breakdowns within India.

SUPPORTING POINTS

- Not available

Distribution & Service Network — scored 80/100

> The evaluation rests on the company's confirmed operational presence localized within the state of Maharashtra. Based on primary research, the company has not disclosed the full extent of its domestic branch offices, regional distributor networks, or service response times across India.

SUPPORTING POINTS

- Pollution Protection Systems Mumbai Pvt. Ltd. operates its registered office and manufacturing works in Thane, Maharashtra, with an additional facility located in the Rabale locality of Navi Mumbai.

Regulatory Approvals & Tenders — scored 98/100

> The rating is based on verified Indian regulatory certifications from PESO and CIMFR for the company's hazardous area gas detectors, combined with documented public tender wins for industrial safety equipment. Based on primary research, the company has not disclosed the complete, multi-year volume or total aggregate valuation of its private-sector tender contracts.

SUPPORTING POINTS

- Pollution Protection Systems Mumbai Pvt. Ltd. possesses Indian regulatory approvals from the Petroleum and Explosives Safety Organisation (PESO) and the Central Institute of Mining and Fuel Research (CIMFR), alongside global certifications like ATEX, IECEx, and SIL-2, for its proprietary gas monitoring and flame detection hardware such as the "gaZsens Series".
- The company successfully won a government tender through a public bid opening on June 5, 2024, securing a contract as the lowest (L1) bidder for gas detection systems under tender reference GEM/2024/B/5007990.

Local Manufacturing & Partnerships — scored 84/100

> The assessment rests on the company’s confirmed domestic presence via an operational production facility in Maharashtra alongside established international distribution collaborations. Based on primary research, the company has not disclosed the precise details of domestic sourcing percentages, joint venture capital structures, or localized technology transfers.

SUPPORTING POINTS

- The company reports operating an integrated manufacturing facility and works located at Devendra Industrial Estate, Lokmanya Nagar in Thane, Maharashtra.
- The company reports active trading, distribution, and technology collaboration partnerships with international brands including SENSIT Technologies (US) and E2S Warning Signals (UK).

HNL Systems Pvt. Ltd. HNL Systems ● Trailblazers X 90 Y 86 Overall 88

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 87/100

> The rating rests on the company's established ability to manufacture multi-gas monitors and flame detectors utilizing a broad portfolio of 12 distinct sensor technologies across multiple measurement ranges (ppb, ppm, %vol, and %LEL). Based on primary research, the company has not disclosed the precise response times (T90 limits) and exact detection accuracy thresholds across its entire core industrial product line.

SUPPORTING POINTS

- The company reports employing 12 different sensor technologies within its product line, including electrochemical, catalytic pellistor/combustion, non-dispersive infrared (NDIR), photoionization detector (PID), semiconductor, thermal conductivity (TCD), and ultraviolet (UV).
- The company reports that its portable gas detectors, such as the Multi Gas Detector GD-102, can measure parts-per-billion (ppb), parts-per-million (ppm), and percentage levels of gases, incorporating poison-resistant pellistor or integrally mounted electrochemical sensors with cross-interference filters.

Product Portfolio Breadth — scored 85/100

> The assessment rests on the company's documented product lines covering portable detectors, fixed transmitters, control panels, and multi-spectrum flame detectors across several sensor technologies. Based on primary research, the company has not disclosed third-party verification or independent technical audits confirming the market-wide relative deployment or performance metrics of its portfolio breadth.

SUPPORTING POINTS

- The company reports offering portable gas detectors (such as the GD-103, GD-102, and GD-101), fixed monitoring systems (GD-250 and GD-200), receiver controllers (RC series including RC-200 and RC-250), and various IR, UV, and UV/IR series flame detectors.
- The company reports utilizing 12 distinct sensor technologies (including combustible gas, infrared point, ultrasonic, photoionization, semiconductor, electrochemical, and galvanic cell) to target applications across industries like oil and gas, power, steel, and petrochemicals.

Intrinsic Safety & Certification — scored 88/100

> The rating rests on the company's verified compliance with key Indian safety approvals (PESO, CMRI, and DGMS) and self-reported adherence to generic international IEC and ATEX metrics across its portfolio. Based on primary research, the company has not disclosed specific third-party certificate registration numbers or direct documentation for localized global standards like ATEX and IECEx.

SUPPORTING POINTS

- HNL Systems Pvt. Ltd. holds regulatory and regional approvals including PESO, CMRI, and DGMS certifications for its hazardous area safety products.
- The company reports compliance with CE marking and international IS/IEC standards across its gas monitoring and flame detection product ranges.

Environmental Durability & Reliability — scored 96/100

> The rating rests on the company's baseline environmental parameters, specifically its deployment of standard IP-rated housings and a 0°C to 55°C operating range across core gas and flame detection models. Based on primary research, the company has not disclosed mean time between failures (MTBF) data or independent field reliability reports.

SUPPORTING POINTS

- The company reports that its fixed gas monitoring systems, gas analyzers, and UV/IR flame detectors utilize enclosures rated at IP65, IP66, or IP67 to provide dust-tight and water-resistant protection in hazardous industrial atmospheres.
- Technical specifications provided by the company indicate standard operational temperature boundaries typically spanning from 0°C to 55°C for its primary online gas analyzers and single/multi-gas monitoring systems.

Integration & Smart Features — scored 92/100

> The rating rests on the company's established, self-reported capabilities to offer industry-standard wired integrations (4-20mA, Modbus, HART) and localized wireless hardware controllers. Based on primary research, the company has not disclosed advanced IoT cloud platform features, integrated edge analytics software, or independent third-party smart feature verifications.

SUPPORTING POINTS

- The company reports that its Field Gas Transmitter TX-250 and various gas detectors feature standard 4-20mA analog signals alongside options for RS485 Modbus and HART communication protocols to interface with control room systems like PLC and SCADA.
- The company reports that its RC-253 Wireless Repeater functions as an innovative wireless gas detection system capable of continuously scanning, configuring, and logging real-time data from up to 32 wireless transmitters.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 100/100

> We verified that the company holds commercial accounts with major industrial, oil and gas, and defense entities in India, but specific deployment volumes or long-term operational counts were not discovered. Based on primary research, the company has not disclosed its precise active installed base or quantitative unit metrics for these projects.

SUPPORTING POINTS

- The company reports long-term clients that include high-profile Indian public and private sector organisations such as HPCL, BPCL, BCCL, CMPDI, BARC, ONGC, the Indian Navy, the Indian Army, NTPC, IOCL, Reliance, Hindustan Unilever, Coal India, and TATA.

Revenue & Growth in India — scored 69/100

> Based on primary research, the company has not disclosed its official financial reports, historical revenue breakdown, or market share data for public verification.

SUPPORTING POINTS

- Not available

Distribution & Service Network — scored 86/100

> The evaluation rests on confirmed physical locations restricted to the Mumbai metropolitan region, along with self-reported capabilities to service clients across India and select international regions. Based on primary research, the company has not disclosed the specific size of its partner network, exact distributor counts, regional service center offices, or standard service response times.

SUPPORTING POINTS

- The company reports its corporate office and manufacturing facilities are located at the Laxmi Industrial Estate in Andheri West, Mumbai, with an additional operations presence in Rabale MIDC, Navi Mumbai.
- The company reports that it supplies products to clients across India, the Middle East, South-East Asia, and Africa, and offers "PAN INDIA" calibration services for gas detectors.

Regulatory Approvals & Tenders — scored 93/100

> The evaluation rests on the company’s documented compliance with Indian industrial safety standards, including PESO testing and CMRI certifications, alongside its verified public sector tender wins with entities like NPCIL. Based on primary research, the company has not disclosed comprehensive records of its private commercial tender volumes or international regulatory body approvals.

SUPPORTING POINTS

- HNL Systems Private Limited holds ISO 9001:2015 certification, and its gas monitoring devices and fixed detectors feature PESO, CIMFR/CMRI, and DGMS compliance or testing certifications for hazardous environments.
- HNL Systems Private Limited was awarded a ₹5.92 Lakh public tender published by the Nuclear Power Corporation of India Limited (NPCIL) for the supply and calibration of portable hydrogen analyzers.

Local Manufacturing & Partnerships — scored 80/100

> The rating rests on the company's active self-reported operation of a physical production facility in Navi Mumbai and its aligning of electronic gas safety instrument assembly with national localization initiatives. Based on primary research, the company has not disclosed third-party verification of its local manufacturing capacity or specific strategic technology transfer partnerships.

SUPPORTING POINTS

- The company reports operating a dedicated manufacturing factory at Plot No. A-59, TTC Industrial Area, MIDC Khairane, Kopar-Khairane, Navi Mumbai, India.
- The company reports manufacturing its line of gas detecting products, flame detectors, and analyzers under the 'Make In India' initiative with an infrastructure that incorporates research and development (R&D) units.

Respo Products Respo Trailblazers X 97 Y 79 Overall 88

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 90/100

> The rating rests on detailed technical specifications disclosed for the company's proprietary fixed and wireless gas detection lines, which demonstrate precise measurement capabilities (ppm, LEL, %vol) across multiple advanced sensor types. Based on primary research, the company has not disclosed independent validation or third-party test data verifying these response times and sensor ranges under external testing protocols.

SUPPORTING POINTS

- The company reports that its Respo BS-03 Fixed Gas Detector uses dual-beam non-dispersive infrared (NDIR), catalytic combustion, electrochemical, and photoionization detector (PID) smart sensor modules to detect target gases like carbon dioxide (0–6000 ppm, response time ≤3 seconds), butane (0–100% LEL, response time ≤3 seconds via NDIR), and carbon monoxide (0–2000 ppm, response time ≤10 seconds).
- The company reports that its remoREAD battery-powered wireless gas detector utilizes NDIR and molecular property spectrometer (MPS) technology for methane/combustibles with a response time of less than 5 seconds full scale.

Product Portfolio Breadth — scored 100/100

> The evaluation rests on the company's verified capability as a system integrator and manufacturer providing a full spectrum of gas detection tubes, portable personal monitors, fixed hazardous area systems, control panels, and specialized visual flame detectors. Based on primary research, the company has not disclosed the precise independent third-party market validation or certification breakdowns for its entire combined proprietary manufacturing line compared to its third-party brand distributions.

SUPPORTING POINTS

- Respo Products reports a combined portfolio of proprietary and distributed brands including Senko, Crowcon, Micropack, Gastec, and GDI, offering single and multi-gas portable detectors, VOC monitors, fixed gas systems like the BS-03 and remoREAD wireless detector, GasGuard M16 control panels, and the Micropack FDS300 series visual or radiant flame detectors.
- The company reports a technical specification where its distributed Crowcon Detective IQ area monitor supports real-time multi-gas detection for up to 7 gases simultaneously, integrating interchangeable sensor slots for electrochemical, PID, infrared, MPS, pellistor, and oxygen sensors.

Intrinsic Safety & Certification — scored 98/100

> We verified that the company maintains explicit ATEX and SIL2 third-party certifications across its proprietary hazardous area fixed and wireless gas detection lines. Based on primary research, the company has not disclosed the independent registration numbers or public download links for its specific PESO and BIS certifications.

SUPPORTING POINTS

- Respo Products manufactures the BS-03 fixed hazardous area gas detector, which holds ATEX Ex dB IIC T6 Gb compliance and has been officially tested under EN 60079 and EN 50270 standards.
- The company's battery-powered remoREAD wireless gas detector is engineered for natural gas environments and features ATEX (Ex db IIC Gb T5) certification alongside SIL2 functional safety compliance according to IEC 61508:2010 standards.

Environmental Durability & Reliability — scored 98/100

> The evaluation rests on the company's published technical specifications for its wireless and handheld detector models, which confirm robust ingress protection ratings and wide operating temperature ranges. Based on primary research, the company has not disclosed MTBF data or field reliability reports.

SUPPORTING POINTS

- The company reports that its remoREAD Wireless Gas Detection System features IP68 ingress protection, an operating temperature range of -40°C to +70°C, and carries ATEX and SIL2 approvals.
- The company reports that its HRLD-300/600 Handheld Laser Methane Leak Detector provides IP66 protection, operates within a temperature range of -20°C to +50°C, and features ATEX explosion-proof certification.

Integration & Smart Features — scored 100/100

> The rating rests on the company's verified capability to provide wireless connectivity and multi-protocol control system integration. Based on primary research, the company has not disclosed independent validation or third-party case studies quantifying data analytics performance.

SUPPORTING POINTS

- The company reports that its remoREAD Wireless Gas Detection System transmits real-time readings, battery status, and alarm alerts to a corresponding gateway, which can further push data to a subscription-based cloud, SMS, and email notifications.
- The company reports that its installation and commissioning services validate 4–20 mA, Modbus, HART, and relay signals to support integration with PLC, DCS, and SCADA systems.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 100/100

> We verified that the company's active accounts include major Indian public sector undertakings and oil & gas enterprises based on their corporate client declarations. Based on primary research, the company has not disclosed granular details regarding its active installation base, specific project locations, or total installed unit counts across India.

SUPPORTING POINTS

- The company reports providing gas detection solutions to a client base that includes public sector undertakings (PSUs) and private sector companies such as HP Oil Gas, BPCL (including BPCL Tirupati), ONGC, and IOCL.
- The company reports metrics of over 5 completed projects and more than 25 satisfied customers for its industrial safety and gas detection systems.

Revenue & Growth in India — scored 65/100

> Based on primary research, the company has not disclosed its sales revenue or annual growth rates in India.

SUPPORTING POINTS

- Not available

Distribution & Service Network — scored 70/100

> We verified that the company's confirmed physical network consists of its single headquarters and production facility in Agra. Based on primary research, the company has not disclosed the number of its branch offices, partner distributors, service centers, or specific service response times across India.

SUPPORTING POINTS

- Respo Products operates from a registered office and a manufacturing factory location in Agra, Uttar Pradesh, India.

Regulatory Approvals & Tenders — scored 75/100

> We verified that the company publicly reports pursuing certifications such as PESO and BIS, but it lacks third-party documentation of active approval status or successful government tender wins. Based on primary research, the company has not disclosed formal registration tracking, current PESO certificate numbers, or awarded tender metrics.

SUPPORTING POINTS

- The company reports that it is actively pursuing local and global certifications, including PESO, BIS, IECEx, ATEX, and SIL2, to strengthen its status as an export-ready Indian manufacturer.

Local Manufacturing & Partnerships — scored 84/100

> The rating rests on the company's verified operational infrastructure in Agra, India, and its active product integration partnerships with prominent global gas detection hardware brands. Based on primary research, the company has not disclosed localized technology transfer frameworks or joint-venture equity configurations backing these foreign product distribution partnerships.

SUPPORTING POINTS

- The company reports operating a modern manufacturing facility located at A-29, EPIP, UPSIDA Industrial Area, Sikandra, Agra 282 007, India, supporting its transition into full-scale "Make in India" manufacturing for fixed and portable gas detection systems.
- The company reports supplying gas and flame detection equipment from international partner brands, listing products from Senko, Crowcon, Micropack, and Gastec under its portfolio.

Unitech Technocrats Private Limited Unitech ● Trailblazers X 86 Y 82 Overall 84

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 81/100

> The rating rests on the company's self-reported technical specifications for its specific gas detection product lines, including target ranges, response times, and sensor methodologies. Based on primary research, the company has not disclosed independent, third-party validation for these specific performance limits and response times across its portfolio.

SUPPORTING POINTS

- The company reports that its UT-1703-7 LEL Gas Detector utilizes electrochemical and pellistor sensors to achieve a wide sensing range from 0 to 100% LEL.
- The company reports that its UT-1702-2 CO2 Smart Gas Detector incorporates electrochemical and NDIR technologies to monitor carbon dioxide across a range of 0 to 50,000 ppm with a response time of less than 20 seconds and an accuracy of ±2% full scale.

Product Portfolio Breadth — scored 80/100

> The evaluation rests on the company's documented range of both fixed and portable safety devices alongside dedicated UV/IR optical flame detectors and flameproof controllers. Based on primary research, the company has not disclosed third-party verification details regarding global market adoption metrics or independent comparative performance benchmarks for its portfolio breadth.

SUPPORTING POINTS

- The company reports offering a product portfolio that includes fixed gas detectors (such as the UT-1700 series), portable gas detectors, wireless gas transmitters, multichannel gas monitoring displays, flameproof control panels, and the UT-FD UV + Triple IR flame detector.
- The company reports that its gas detection systems cover oxygen, lower explosive limits (LEL) for combustible gases, volatile organic compounds (VOCs), and toxic gases including carbon monoxide, sulfur dioxide, ammonia, hydrogen sulfide, hydrogen chloride, chlorine, and carbon dioxide.

Intrinsic Safety & Certification — scored 79/100

> The assessment rests on the company's product literature claiming explosion-proof enclosure certifications through regional Indian bodies such as CIMFR and PESO, alongside ATEX(CE) compliance for hazardous area applications. Based on primary research, the company has not disclosed independent third-party certificate registration numbers or verifiable documentation from international compliance databases.

SUPPORTING POINTS

- The company reports that its gas detector enclosures feature CIMFR and PESO approvals for gas groups IIA, IIB, and IIC.
- Product datasheets published by the company indicate that its smart gas detectors feature ATEX(CE) certifications.

Environmental Durability & Reliability — scored 96/100

> The assessment rests on the company's detailed technical specifications confirming specific ingress protection ratings and standard industrial operating temperature envelopes for its gas and flame detection instruments. Based on primary research, the company has not disclosed third-party field reliability records or mean time between failures (MTBF) data.

SUPPORTING POINTS

- Unitech Technocrats Private Limited reports that its carbon dioxide (UT-1702-2) and oxygen (UT-1701-1) gas detectors are designed with an IP65, IP66, or IP67 protection class, operating temperature ranges between -10°C to 50°C or -15°C to 60°C, and are housed in CMRI-approved enclosures for industrial hazardous environments.
- The company reports that its UV and Triple IR flame detectors (Model UT-FD) are engineered with an IP65 and IP68 protection class, a cast aluminium alloy LM-6 enclosure, and can operate in temperatures ranging from -20°C to 85°C with less than 95% non-condensing humidity.

Integration & Smart Features — scored 92/100

> The rating rests on the company's self-reported capabilities to design control panels that integrate wireless gas detection hardware with PLC/SCADA systems using standard protocols like Modbus and RF communication. Based on primary research, the company has not disclosed third-party verification, detailed data analytics software platforms, or broader IoT cloud compatibility specific to its gas and flame safety systems.

SUPPORTING POINTS

- The company reports providing custom PLC-based control panels designed for trouble-free, seamless integration with wireless gas detectors and transmitters, alongside multichannel gas monitoring displays.
- The company reports that its automation systems support standard protocols including Modbus TCP/IP, Modbus RTU, and RF (Radio Frequency) communication via its Smart Wireless Gateways.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 72/100

> Based on primary research, the company has not disclosed the number of active installations or names of major customers in India.

SUPPORTING POINTS

- Not available

Revenue & Growth in India — scored 96/100

> The rating is based on the company's confirmed positive year-over-year revenue and net sales growth metrics through corporate tracking databases. Based on primary research, the company has not disclosed its exact revenue figures specifically derived from its gas and flame detection product segment within the Indian market.

SUPPORTING POINTS

- Unitech Technocrats Private Limited reported a net sales revenue increase of 7.92% in 2025.
- The company's operating revenue range is between INR 1 crore and 100 crore for the financial year ending on 31 March, 2024, with its EBITDA increasing by 61.73% over the previous year.

Distribution & Service Network — scored 77/100

> The evaluation rests on the company’s documented presence at a centralized corporate office and manufacturing site in Vadodara, Gujarat, with no verifiable regional branch offices or external partner networks. Based on primary research, the company has not disclosed the count or location of independent distributors, dedicated service centers, or certified support engineers across India.

SUPPORTING POINTS

- The company reports its corporate office and R&D facilities are situated at a single principal location at Plot No. 13, Sevasi TP-2, Sevasi Canal Road, Sevasi, Vadodara, Gujarat.

Regulatory Approvals & Tenders — scored 87/100

> The rating rests on the company's self-reported claim of holding PESO certification for its smart gas detector module. Based on primary research, the company has not disclosed independent verification of its regulatory licenses or official listings of government/private tender wins in the gas and flame detection market.

SUPPORTING POINTS

- The company manufactures a smart gas detector module that it claims is PESO-certified with PPM detection and 4-20mA output.

Local Manufacturing & Partnerships — scored 80/100

> The rating rests on the company's verified footprint as a domestic manufacturer of gas detectors and industrial process instrumentation within its own facility in Vadodara. Based on primary research, the company has not disclosed strategic joint ventures or technology transfers with international partners for its gas and flame detection portfolio.

SUPPORTING POINTS

- The company reports operating an urban manufacturing facility spanning 10,000 square feet located in Vadodara, Gujarat, where it performs in-house manufacturing for over 160 products, including gas monitoring solutions like LEL, oxygen, and toxic gas detectors.
- The company reports that its facilities include an in-house National Accreditation Board for Testing and Calibration Laboratories (NABL) accredited laboratory and hold an ISO 9001:2015 certification for its instrumentation and automation systems.

Arka Instruments Incorporation Arka ● Trailblazers X 87 Y 74 Overall 80

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 81/100

> The rating rests on the company's disclosed self-reported specifications for its portable and fixed online gas detection systems across various sensor types and targets. Based on primary research, the company has not disclosed third-party performance validations or independent certifications verifying these accuracy and response capabilities under industrial conditions.

SUPPORTING POINTS

- The company reports that its AI-95H Online Gas Detector utilizes catalytic, electrochemical, photoionization detection (PID), or infrared principles to detect over 40 toxic, combustible, and oxygen gases with a precision of ≤±3% F.S..
- The company reports that its AI-4A Portable Multi-Gas Detector achieves a 10-second response time and measures ranges of 0–1000ppm for CO, 0–100ppm for H2S, 0–30%vol for O2, and 0–100%LEL for combustible gases using plug-in electrochemical and catalytic combustion sensors.

Product Portfolio Breadth — scored 80/100

> The evaluation rests on the company's documented product portfolio spanning portable detectors, fixed online monitoring systems, and dedicated multi-channel touch-screen control panels covering a broad spectrum of industrial gases. Based on primary research, the company has not disclosed the design or manufacture of dedicated optical flame detectors within its safety instrumentation lines.

SUPPORTING POINTS

- The company reports offering portable gas detectors (including models like AI-4A, AI-4AS, AI-1200H-S, AI-2000 M4, and AI-90E), fixed gas detectors (such as the AI-100 series and AI-95H online detector), and multi-channel gas monitoring control panels (including the AI-50 touch screen system).
- The company reports that its gas detection devices can monitor over 40 types of toxic, combustible, and oxygen gases, including carbon monoxide, hydrogen sulfide, methane, ammonia, volatile organic compounds (VOCs), and chlorine.

Intrinsic Safety & Certification — scored 81/100

> The assessment rests on the company's self-reported technical documentation claiming internal protection ratings, flameproof enclosure certifications through India's CIMFR, and cross-listings for PESO-approved systems. Based on primary research, the company has not disclosed third-party verification documents or direct certificate registry details confirming current ATEX, IECEx, or BIS compliance for its proprietary line of gas detectors.

SUPPORTING POINTS

- The company reports that its process indicators and temperature controllers utilize flameproof enclosures certified by CIMFR for Gas Groups I, IIA, and IIB, and its tanker earth monitoring systems carry CIMFR / PESO certification for explosion-proof (Ex d) and intrinsically safe circuits.
- The company's fixed gas detectors (such as the AI-95H and VOC models) are designed with explosion-proof ratings specified as Ex d II CT6 with an IP65 protection class.

Environmental Durability & Reliability — scored 95/100

> The assessment rests on the company's published technical specifications detailing IP65 ingress protection, explosion-proof housing certifications, and operational temperature ranges across its product portfolio. Based on primary research, the company has not disclosed mean time between failures (MTBF) data, quantified false alarm rates, or independent field reliability reports.

SUPPORTING POINTS

- The company reports that its Arka Instruments AI-95H Online Gas Detector features an IP65-rated, water-, dust-, and explosion-proof (Ex dIICT6) die-cast aluminum or stainless steel case, supported by optional high-temperature probes rated up to 1200°C.
- The company reports that its Arka Instruments VOC / TVOC Fixed Gas Detector is rated for an operating temperature range of -20°C to 55°C and humidity levels of 0% to 95% RH under an IP65 protection class.

Integration & Smart Features — scored 96/100

> The rating rests on the company's verified capability to offer Modbus, RS485, and optional wireless outputs for its fixed gas detectors alongside dedicated IoT data-logging hardware. Based on primary research, the company has not disclosed independent third-party certifications for its smart software ecosystem or advanced automated analytics capabilities.

SUPPORTING POINTS

- The company reports that its AI-100 and AI-95H fixed gas detectors support multiple signal outputs including 4-20mA current, RS485 digital (Modbus), relay signals, and optional wireless digital signals, allowing integration with external controllers and computer software for real-time monitoring and data transmission.
- The company reports that its IoT DataLogger features WiFi and 4G SIM connectivity, supports Modbus protocols via RS485 input, and integrates with the Arka WebApp to provide dashboards, data exports, and email/SMS alerts.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 65/100

> We verified that no specific customer names, project references, or unit installation numbers are available in public company disclosures or independent deployment records for the company's gas detection systems. Based on primary research, the company has not disclosed its active installed base counts or key account lists in India.

SUPPORTING POINTS

- Not available

Revenue & Growth in India — scored 91/100

> The rating rests on registered regulatory tracking showing that the company operates as a small-scale entity generating under ₹10 Cr in annual Indian revenue. Based on primary research, the company has not disclosed audited revenue breakdowns, specific year-over-year commercial sales figures, or verified market share metrics within the industrial gas and flame detection segment.

SUPPORTING POINTS

- The company's entity Arka Instruments LLP (incorporated in September 2023) generated less than ₹10 Cr in revenue for the financial year ending on March 31, 2025.

Distribution & Service Network — scored 70/100

> The evaluation rests on the company's single documented operational presence in Hyderabad, Telangana. Based on primary research, the company has not disclosed any expansive regional sales branches, authorized third-party distributor networks, or service response times across India.

SUPPORTING POINTS

- The company reports a primary office location in Hyderabad, Telangana, which handles the manufacturing, supplying, and exporting of its gas detectors and monitoring systems.

Regulatory Approvals & Tenders — scored 68/100

> Based on primary research, the company has not disclosed its regulatory compliance certificates or contract win history for its gas detection product lines.

SUPPORTING POINTS

- Not available

Local Manufacturing & Partnerships — scored 74/100

> The rating rests on the company's self-reported operational infrastructure for manufacturing and quality testing gas detection devices in Hyderabad, India. Based on primary research, the company has not disclosed third-party verification of its local manufacturing capacity or details regarding joint ventures and technology transfer partnerships in India.

SUPPORTING POINTS

- The company reports operating an infrastructure facility in Hyderabad, Telangana, divided into manufacturing, quality testing, research & development, warehouse, transportation, and packaging units to produce its gas detectors.
- Arka Instruments Incorporation maintains a localized presence through its registered facility in Rangareddy, Hyderabad, which caters to supplying and exporting gas detectors.

Mukta Instruments ● Evolving Players

X 83 Y 79 Overall 81

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 79/100

> The evaluation rests on the company's self-reported technical specifications for portable and fixed detection devices that include infrared, catalytic, and electrochemical sensor technologies. Based on primary research, the company has not disclosed independent third-party performance verifications or comprehensive detection limits across all supported gases.

SUPPORTING POINTS

- The company reports its fixed gas detectors utilize catalytic combustion and infrared sensors, achieving a response time of less than 20 seconds for units like the MI-95H and measuring ranges such as 0–100 PPM for hexane.
- The company reports its portable gas detectors, such as the MI-20B and MI-2000, utilize electrochemical and infrared principles to measure units in PPM, mg/m³, and %VOL with a response time under 30 seconds.

Product Portfolio Breadth — scored 76/100

> The rating rests on the company's comprehensive self-reported collection of portable and fixed gas detection equipment covering over 20 hazard gases and multi-channel control interfaces. Based on primary research, the company has not disclosed independent third-party verification of its portfolio breadth or the availability of dedicated industrial flame detectors.

SUPPORTING POINTS

- The company reports offering an extensive gas detection portfolio that includes portable gas detectors, fixed gas detectors, gas analyzers, laser methane mini detectors, and multi-channel control panels (8-channel, 16-channel, and 32-channel models) manufactured in India.
- The company reports that its product portfolio covers single and multi-gas detection (up to 12-in-1 configurations) for a wide range of gases including O3, SF6, H2, He, O2, LPG, VOC, H2S, hexane, methanol, NH3, CO, CO2, SO2, Cl2, CH4, HCN, HF, PH3, ETO, HCl, NO, N2, and LEL.

Intrinsic Safety & Certification — scored 77/100

> We verified that Mukta Instruments claims compliance with global safety standards like ATEX and IECEx for select fixed and portable gas systems, as well as Chinese GB industrial standards. Based on primary research, the company has not disclosed localized third-party certificates or official approvals from Indian authorities such as PESO or BIS.

SUPPORTING POINTS

- The company reports that its MI-100 Fixed Gas Detector is built with an explosion-proof grade of Ex Db IIC T6 Gb and Ex Db Ia IIC T6 Gb under both ATEX and IECEx standards.
- The company reports that its MI-MiniMeta O2 Gas Detector complies with Chinese explosion-proof guidelines and executive standards GB 12358-2006, GB 3836.1-2010, and GB 3836.4-2010.

Environmental Durability & Reliability — scored 93/100

> The evaluation rests on self-reported technical specifications detailing ingress protection ratings and operational limits across the company's product line. Based on primary research, the company has not disclosed third-party durability certifications, MTBF metrics, or field reliability data.

SUPPORTING POINTS

- The company reports an operating temperature range of -40℃ to 70℃ for the catalytic sensor and an IP66/67 protection grade for its MI-100 Fixed Gas Detector.
- The company reports an IP68 protection class, a working temperature range of -20℃ to 55℃, and a working humidity range of 0–95% RH for its MI-MiniMeta series detectors.

Integration & Smart Features — scored 92/100

> The evaluation rests on the company's self-reported capabilities to offer both wired communication protocols (4-20mA, RS485) and diverse wireless expansion modules (HART, LoRa, GPRS, 4G) for remote data collection across its fixed gas detection lineup. Based on primary research, the company has not disclosed specific third-party validation or verified case studies illustrating large-scale integration into SCADA or PLC industrial control architectures.

SUPPORTING POINTS

- The company reports that its MI-100 Fixed Gas Detector supports standard 4-20mA, RS485, and 1-5V communication protocols, with optional integration capabilities for HART, LoRa, GPRS, and 4G wireless networks.
- The company reports that its MI-98H Fixed Gas Detector features Chinese and English language support along with dedicated PC software enabling automated data downloading and analysis.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 72/100

> We verified that the company's client indicators are limited to individual buyer reviews and generic professional roles. Based on primary research, the company has not disclosed major customer names, verified industrial project references, or installed unit counts within oil & gas, petrochemical, or manufacturing facilities.

SUPPORTING POINTS

- The company reports customer feedback from individual roles, including a maintenance supervisor in Bengaluru, an industrial service contractor in Maharashtra, and a safety officer in Hyderabad.
- Client transaction and satisfaction metrics indicate a pan-India user distribution across cities such as Chennai, Coimbatore, Kochi, New Delhi, Bengaluru, and Nagpur.

Revenue & Growth in India — scored 91/100

> The assessment rests on the company's self-reported overall annual turnover ranges found across its business listings and digital catalogs. Based on primary research, the company has not disclosed audited financial reports, segmented revenue for its gas and flame detection division, or historical year-over-year commercial growth rates in the Indian market.

SUPPORTING POINTS

- The company reports an annual turnover of ₹10 Crores on its official export profile and industrial trade directory listings.
- The company's self-reported commercial profile on industrial marketplaces lists a lower annual GST turnover bracket of ₹1.5 Crores to ₹5 Crores.

Distribution & Service Network — scored 93/100

> The rating rests on the fact that Mukta Instruments operates out of a single confirmed office location in Bengaluru while utilising a self-stated channel partner network for broader product delivery. Based on primary research, the company has not disclosed the exact number of its distributors, physical service center locations, or quantified service response times across India.

SUPPORTING POINTS

- The company reports having a network of channel partners to deliver its products, but officially lists its primary physical operation at a single location in Bengaluru, Karnataka.

Regulatory Approvals & Tenders — scored 68/100

> Based on primary research, the company has not disclosed its regulatory approvals and tender history.

SUPPORTING POINTS

- Not available

Local Manufacturing & Partnerships — scored 72/100

> The rating rests on the company's self-reported operational presence as a manufacturer and supplier based in Bengaluru. Based on primary research, the company has not disclosed independent third-party verification of its local manufacturing facilities or any strategic partnerships for technology transfer in India.

SUPPORTING POINTS

- The company reports operating as a manufacturer, exporter, supplier, and trader of industrial instruments, including fixed and portable gas detectors, from its facility in Bengaluru, Karnataka.

Hythane Technologies Private Limited

Hythane Technologies · Evolving Players · X 80 · Y 80 · Overall 80

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 90/100

> The rating rests on the company's confirmed product architecture for its e-Sniffer platform, which displays an established capability across multiple major industrial sensor technologies and explicit detection limits ranging from parts-per-million to percent-by-volume. Based on primary research, the company has not disclosed quantified response times (such as T90 limits) or exact accuracy tolerances.

SUPPORTING POINTS

- The company reports that its core product, the e-Sniffer industrial gas detector, covers 17 gas types across 33 variants—including hydrocarbons, hydrogen, and toxic and flammable gases—with configurations stretching from PPM to %-volume ranges.
- The company reports that its core technology stack incorporates metal oxide semiconductor (MOS), catalytic, and electrochemical sensor types, utilizing end-to-end in-house design from nanomaterial thin-film synthesis to hardware firmware.

Product Portfolio Breadth — scored 70/100

> The rating rests on verified procurement records and company documentation confirming the manufacture of its fixed "e-Sniffer" platform for multi-gas monitoring variants, particularly hydrogen and hydrogen sulfide sensors. Based on primary research, the company has not disclosed any commercial portfolio covering portable gas detectors, standalone flame detectors, or dedicated control panels.

SUPPORTING POINTS

- The company reports offering the "e-Sniffer" industrial gas detector, a micro-controller-based fixed gas detection platform capable of detecting 17 gas types across 33 configuration variants using metal oxide semiconductor (MOS), catalytic, and electrochemical sensors.
- A public tender document from 2026 verifies the industrial supply of the company's "e-Sniffer" model HS20/50XX, which operates as a MOS sensor-based fixed hydrogen sulfide gas detector.

Intrinsic Safety & Certification — scored 81/100

> The rating rests on the company's detailed public listings of applicable local Indian regulatory certifications alongside statement disclosures that full international ATEX and IECEx compliance verifications are in progress for its core product line. Based on primary research, the company has not disclosed the finalized certificates or official registration numbers from independent global testing bodies.

SUPPORTING POINTS

- The company reports that its e-Sniffer industrial gas detector platform lists approvals or compliance alignments including DAE/BARC acceptance, IS/IEC Ex, PESO, and CIMFR.
- The company reports that it is currently undergoing formal international IECEx and ATEX certifications for its indigenous line of gas detectors.

Environmental Durability & Reliability — scored 96/100

> The evaluation rests on the company's verified enclosure ingress rating and explosion-proof certifications for hazardous industrial zones. Based on primary research, the company has not disclosed operating temperature ranges, mean time between failures (MTBF) statistics, or field reliability performance reports.

SUPPORTING POINTS

- The company reports that its e-Sniffer industrial gas detector features an IP66 ingress-rated LM-6 cast aluminum enclosure.
- The company reports that the e-Sniffer is Ex-d certified for Zone 1 and Zone 2 hazardous environments under Gas Groups IIA, IIB, and IIC, with approvals from PESO, CIMFR, and IS/IEC Ex.

Integration & Smart Features — scored 65/100

> We verified that the company's product line includes micro-controller-based embedded transmitters, indicating baseline computational capabilities. Based on primary research, the company has not disclosed compatibility with common industrial control systems (such as PLC, SCADA, or IoT frameworks), wireless connectivity, remote monitoring applications, or data analytics capabilities.

SUPPORTING POINTS

- The company reports that its e-Sniffer platform is a micro-controller-based gas-detection transmitter compatible with MOS, Catalytic, and Electrochemical sensing, engineered from the sensor stack up to embedded firmware.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 65/100

> Based on primary research, the company has not disclosed its customer names, project references, or installed unit counts.

SUPPORTING POINTS

- Not available

Revenue & Growth in India — scored 85/100

> The rating is based on the company's early-stage status as an indigenous developer of gas detection devices with baseline corporate registration details verified through public filings. Based on primary research, the company has not disclosed broken-down sales revenue figures, year-over-year commercial growth rates, or independent market share data in India.

SUPPORTING POINTS

- Hythane Technologies Private Limited reports designing and manufacturing the e-Sniffer industrial gas detector featuring in-house nanomaterials, hardware, and firmware.
- Ministry of Corporate Affairs records indicate that the company was incorporated on March 30, 2024, and its authorized and paid-up share capital are both ₹1 Lakh.

Distribution & Service Network — scored 72/100

> We verified that the company's confirmed operations are restricted to its headquarters and corporate office setup in Maharashtra. Based on primary research, the company has not disclosed the number of additional regional offices, external distributors, partner networks, or standard service response times across India.

SUPPORTING POINTS

- The company reports its design and manufacturing headquarters is located in Navi Mumbai, India, with an additional corporate office located in Thane, Maharashtra.

Regulatory Approvals & Tenders — scored 91/100

> We verified that the company holds crucial local compliance certifications including PESO and CIMFR approvals, alongside participation in public sector procurement tenders. Based on primary research, the company has not disclosed the volume or total monetary value of its secured government and private contracts.

SUPPORTING POINTS

- The company reports that its micro-controller-based transmitter for industrial gas detection holds IS/IEC Ex, PESO, and CIMFR approvals.
- Procurement records indicate a public tender opening on 24th March, 2026, for the supply of a "M/s Hythane Make E-sniffer Model Hs20/50xx" MOS sensor-based fixed hydrogen sulfide gas detector under the Prime Minister of India organisation category.

Local Manufacturing & Partnerships — scored 88/100

> We verified that the company has established domestic manufacturing facilities and strategic institutional technology collaborations for its gas detection products. Based on primary research, the company has not disclosed its exact local sourcing percentages or localized production volume metrics.

SUPPORTING POINTS

- The company maintains its design and manufacturing headquarters in Taloja, Navi Mumbai, producing indigenous gas sensor families right from the sensor element to electronics assembly.
- The company reports a strategic technology partnership with the Bhabha Atomic Research Centre (BARC) and partnered with Process Analytik Instruments Pvt. Ltd. in April 2026 to distribute its advanced thin-film semiconductor gas sensors.

Amgastech Private Limited

Amgastech · Evolving Players · X 83 · Y 77 · Overall 80

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 90/100

> The rating rests on the company's detailed technical specifications for its portable and fixed detection systems, which outline the integration of diverse sensor modalities (catalytic, electrochemical, NDIR) and performance criteria like response times. Based on primary research, the company has not disclosed independent third-party performance verifications or field-testing data.

SUPPORTING POINTS

- The company reports that its portable gas leak detectors and fixed gas transmitters utilize catalytic bead, electrochemical, semiconductor, and non-dispersive infrared (NDIR) sensors to measure gases such as LPG, CNG, methane, carbon monoxide, hydrogen, ammonia, chlorine, hydrogen sulfide, and oxygen.
- The company reports technical specifications including a response time (T90) of 10 seconds or less for its portable leak detectors, an accuracy of ±3% FS with a response time of 15 seconds or less for its lower explosive limit (LEL) gas transmitters, and measuring ranges spanning 0–100% LEL or 0–1000 ppm depending on the gas.

Product Portfolio Breadth — scored 76/100

> The rating rests on the company's verified commercial portfolio of both portable gas detectors and fixed transmitter systems across multiple toxic and combustible gases. Based on primary research, the company has not disclosed any dedicated flame detection models or centralized multi-channel control panels within its product lineup.

SUPPORTING POINTS

- The company reports a portfolio that includes portable gas detectors, fixed gas monitoring systems, multi-gas detectors, gas transmitters, and specialized gas analyzers covering various gases such as oxygen, hydrogen sulfide, carbon monoxide, methane, and ozone.
- The company's catalog includes over 25 listed gas detector products and 9 gas leak detector products alongside fixed systems like the LEL Gas Transmitter.

Intrinsic Safety & Certification — scored 65/100

> The rating rests on the company's detailed product data specifications claiming compliance with standard hazardous area classifications and explosion-proof ratings. Based on primary research, the company has not disclosed formal third-party certificates or independent approvals from regulatory bodies like PESO, BIS, ATEX, or IECEx.

SUPPORTING POINTS

- The company reports that its portable gas detectors are intrinsically safe for use in hazardous areas conforming to Class I, Division I, Group A, B, C & D or Zone 0, Group IIA, IIB & IIC.
- The company notes that its AMG-GT-20FLP gas transmitter is housed in an explosion-proof grade Gas Group IIC flameproof enclosure.

Environmental Durability & Reliability — scored 93/100

> The rating rests on the company's self-reported technical specifications for environmental tolerances and sensor durability across its portable instrument lines. Based on primary research, the company has not disclosed independent third-party field reliability reports or mean time between failures (MTBF) data.

SUPPORTING POINTS

- The company reports an Ingress Protection rating of IP65, an operating temperature range of -20°C to +50°C, and an operating humidity range of 0–95% RH (non-condensing) for its portable gas leak detector.
- The company reports that its portable multi-gas detectors utilize catalytic bead sensor technology to ensure accurate lower explosive limit (LEL) detection and high durability even in harsh environments.

Integration & Smart Features — scored 92/100

> The evaluation rests on the company's verified capability to provide fixed transmitters that utilize ModBus communication and support integration into standard control systems like PLC, DCS, and SCADA. Based on primary research, the company has not disclosed comprehensive software analytics platforms, cloud dashboards, or proprietary IoT remote-monitoring frameworks specific to its gas detection portfolio.

SUPPORTING POINTS

- The company's LEL Gas Transmitter is designed for continuous monitoring and is built for integration with PLC, DCS, and SCADA systems.
- The company's Hydrogen and Combustible Gas Sensor Transmitters feature ModBus protocol connectivity alongside options for wireless configurations.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 67/100

> Based on primary research, the company has not disclosed its customer list, reference projects, or installed base metrics. www.amgastech.in

SUPPORTING POINTS

- Not available

Revenue & Growth in India — scored 91/100

> The evaluation is based on self-reported turnover figures published on the company's product catalog and profile pages, confirming a small commercial footprint. Based on primary research, the company has not disclosed comprehensive financial reports, year-over-year growth rates, or market share estimates.

SUPPORTING POINTS

- The company reports an annual turnover or turnover range of ₹0 - ₹40 Lakhs.

Distribution & Service Network — scored 65/100

> Assessment rests entirely on the lack of verifiable national presence, as we verified only a single primary facility in Maharashtra. Based on primary research, the company has not disclosed the number of active distributors, partner locations, or service turnaround times across India.

SUPPORTING POINTS

- The company reports its primary operational and manufacturing facility is located in Ambarnath, Maharashtra, with up to 10 employees listed at its primary location.

Regulatory Approvals & Tenders — scored 90/100

> The rating rests on the company's verified public tender wins with state enterprises like NTPC and its structured product listings on the official GeM portal. Based on primary research, the company has not disclosed formal, independent third-party PESO certificate registrations held directly under its own corporate manufacturing entity name rather than transactional delivery agreements.

SUPPORTING POINTS

- Amgastech Private Limited has recorded 9 wins across 51 total public tender participations, including a low-bid (L1) win for a ₹3.85 lakh contract with NTPC Limited in Uttar Pradesh.
- The company explicitly commits to providing Petroleum and Explosives Safety Organisation (PESO) approval certificates to buyers at the time of supply for its listed portable multi-gas detectors on the Government e Marketplace (GeM).

Local Manufacturing & Partnerships — scored 74/100

> The rating rests on the company's self-reported operational presence of an active manufacturing facility and warehouse in Maharashtra. Based on primary research, the company has not disclosed any independent third-party verification of its local manufacturing capacity or strategic industry partnerships.

SUPPORTING POINTS

- The company reports operating an office, warehouse, and a local manufacturing unit located in Ambarnath, Maharashtra, India, for the production of gas detectors, gas analyzers, and environmental monitoring equipment.

Tenacious ● Evolving Players X 83 Y 73 Overall 78

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 86/100

> The analysis rests on the company's disclosed product data sheets outlining diverse sensor modalities, resolutions, and response speeds for various industrial gas hazards. Based on primary research, the company has not disclosed independent third-party performance verifications or dedicated industrial flame detection capabilities.

SUPPORTING POINTS

- The company reports utilizing electrochemical, catalytic combustion, semiconductor, and non-dispersive infrared (NDIR) sensor technologies across its fixed gas transmitters to measure hazardous gases in LEL, ppm, and %vol ranges.
- The company reports specific technical specifications including a T90 response time of less than 20 seconds for its electrochemical methane gas detector and a measurement capability up to 30% vol. for its carbon dioxide transmitter series.

Product Portfolio Breadth — scored 65/100

> Our assessment rests on the company's confirmed manufacturing of fixed gas transmitters and combustible leak detectors supporting a diverse array of target gases and sensing principles. Based on primary research, the company has not disclosed the availability of dedicated flame detectors or centralized control panels.

SUPPORTING POINTS

- The company reports a portfolio of gas transmitters for hazardous and normal areas that detect carbon monoxide, carbon dioxide, R32 refrigerant, ammonia, sulfur dioxide, nitrogen dioxide, oxygen, ozone, ethylene oxide, hydrogen, and combustible LEL gases.
- Product specifications verify that the company manufactures fixed gas detectors and combustible leak detectors utilizing electrochemical, semiconductor, and catalytic combustion sensing technologies.

Intrinsic Safety & Certification — scored 81/100

> Our analysis rests on the company's self-reported international compliance ratings and explosion-proof markings for its industrial safety equipment. Based on primary research, the company has not disclosed third-party verification certificates or local Indian hazardous area approvals such as PESO and BIS.

SUPPORTING POINTS

- The company reports ISO, CE, FCC, RoHS, ATEX, CNEX, and SIL3 approvals for its gas detection systems.
- Product specification listings for the company's TGD-500S and TGD-300S fixed gas detectors claim compliance with ATEX, FCC, CE, and RoHS standards.

Environmental Durability & Reliability — scored 93/100

> The assessment rests on the company's confirmed IP67 housing specification and operational intent for high-humidity condensed environments, though these parameters are self-reported. Based on primary research, the company has not disclosed comprehensive third-party verified environmental endurance test data, Mean Time Between Failures (MTBF) figures, or field reliability reports detailing long-term false alarm rates under extreme industrial conditions.

SUPPORTING POINTS

- The company reports that its Fixed Hydrogen Gas Detector features an aluminum enclosure with an IP67 degree of protection.
- The company reports that its TTHCO2 transmitter is designed to provide temperature, humidity, dewpoint, and carbon dioxide monitoring inside condensed environments such as greenhouse and cold storage applications.

Integration & Smart Features — scored 89/100

> The rating rests on the company's self-reported capabilities regarding Modbus-RTU communication protocols, PLC automation compatibility, and its web-based industrial IoT analytics platform. Based on primary research, the company has not disclosed independent third-party verification for its smart features or detailed specifications of its wireless transmission protocols.

SUPPORTING POINTS

- The company reports that its fixed gas transmitters, such as its methane and combustible gas leak detectors, feature standard bus RS485 (Modbus-RTU) simultaneous output and 4-20mA standard signals to ensure compatibility with PLC and DCS systems.
- The company reports providing industrial data analytics services powered by its own IoT hardware and a dedicated web platform to enable real-time monitoring via wired or remote wireless network transmission.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 65/100

> Based on primary research, the company has not disclosed customer names, project references, or installed unit counts in India.

SUPPORTING POINTS

- Not available

Revenue & Growth in India — scored 93/100

> We verified that Tenacious self-reports a total annual turnover between Rs. 1 and 2 Crore for its entire business operation. Based on primary research, the company has not disclosed broken-down sales revenue or year-over-year commercial growth rates specifically for the gas and flame detection market in India.

SUPPORTING POINTS

- The company reports an annual turnover of Rs. 1 - 2 Crore.

Distribution & Service Network — scored 70/100

> Our evaluation rests on the absence of verifiable records regarding the company's logistics footprint or regional service infrastructure in India. Based on primary research, the company has not disclosed the number of its office locations, partner networks, or service response times.

SUPPORTING POINTS

- Not available

Regulatory Approvals & Tenders — scored 65/100

> Based on primary research, the company has not disclosed its compliance certifications, PESO approvals, or won tenders.

SUPPORTING POINTS

- Not available

Local Manufacturing & Partnerships — scored 74/100

> The evaluation rests on the company's self-reported operations as a domestic manufacturer of gas detection instruments in Ahmedabad, India. Based on primary research, the company has not disclosed any joint ventures, technology transfers, or strategic partnerships.

SUPPORTING POINTS

- The company reports operating as a manufacturer, supplier, and exporter of hazardous and normal-area gas transmitters and sensors with production facilities based in Ahmedabad, Gujarat, India, since its establishment in 2022.

Paartha Electronics Gas Kavach ● Evolving Players X 74 Y 72 Overall 73

X — PRODUCT CAPABILITY

Sensor Technology & Detection Range — scored 79/100

> The evaluation rests on the company's self-reported technical specifications for its industrial gas leakage detection systems, which confirm the use of catalytic combustible sensors capable of detecting a specific range of gases starting from 10% LEL with a 5-second response time. Based on primary research, the company has not disclosed independent third-party verification or technical validation for these performance and range metrics.

SUPPORTING POINTS

- The company reports that its industrial gas leakage detection system utilizes multi-point catalytic combustible sensors with a response time of a maximum of 5 seconds.
- The company reports that its industrial gas detection devices offer a detection range starting from 10% of the lower explosive limit (LEL) and upward for specific target gases, which include LPG, LNG, hydrogen, acetylene, and ammonia.

Product Portfolio Breadth — scored 85/100

> The rating rests on the company's confirmed capability to offer a multi-category product portfolio covering fixed industrial panels, portable field detectors, vehicle safety units, and residential alarms across several combustible and toxic gas types. Based on primary research, the company has not disclosed flame detectors or dedicated optical flame detection systems within its product range.

SUPPORTING POINTS

- The company reports offering gas detection solutions categorized into four groups: Industrial & Commercial Systems (including fixed panels like the Panel GV 108 / GV 09 series, multi-channel monitoring, and audio-visual alarms), Portable Gas Detectors (handheld single-gas and multi-gas models with LCD displays), Domestic Gas Detectors (residential LPG and methane detectors), and Auto Gas Alert Systems (for CNG/LPG vehicles and car park CO monitoring).
- The company's products are designed to detect various specific hazardous gases and vapors, including LPG, LNG, CNG, hydrogen, acetylene, ammonia, carbon monoxide, carbon dioxide, and solvent vapors.

Intrinsic Safety & Certification — scored 74/100

> The rating rests on self-reported claims of CMRI and BIS compliance for its gas leakage detection line. Based on primary research, the company has not disclosed independent proof or formal safety approvals (such as ATEX, IECEx, or PESO) required for industrial hazardous area deployment.

SUPPORTING POINTS

- The company reports offering gas leakage detection systems with CMRI and BIS certification.

Environmental Durability & Reliability — scored 65/100

> Based on primary research, the company has not disclosed IP ratings, operating temperature ranges, MTBF data, or field reliability reports.

SUPPORTING POINTS

- Not available

Integration & Smart Features — scored 69/100

> The rating rests on the company's established manufacturing baseline for localized, hardwired multi-channel industrial gas detection panels and automated valve shutoffs. Based on primary research, the company has not disclosed integration support for advanced control systems (such as PLC, SCADA, or IoT setups), remote monitoring software platforms, or data analytics functionality.

SUPPORTING POINTS

- The company reports that its industrial gas monitoring panels (GV 108 / GV 09 series) feature multi-channel monitoring, audio-visual alarms, and an automatic solenoid valve shutoff capability.
- Product specifications from the company indicate its industrial detectors feature hardwired setups spanning up to a maximum distance of 200 meters between the catalytic sensors and the control hub, utilizing basic green and red LED indicators along with a hooter or siren for alert handling.

Y — BUSINESS CAPABILITY

Installed Base & Key Accounts — scored 65/100

> Based on primary research, the company has not disclosed its customer list, active project references, or total installed base metrics.

SUPPORTING POINTS

- Not available

Revenue & Growth in India — scored 69/100

> Based on primary research, the company has not disclosed its financial reports, year-over-year revenue growth, or Indian market share metrics.

SUPPORTING POINTS

- Not available

Distribution & Service Network — scored 65/100

> Based on primary research, the company has not disclosed its distribution network footprint, partner channels, or service response times.

SUPPORTING POINTS

- Not available

Regulatory Approvals & Tenders — scored 89/100

> The evaluation rests on an established public sector contract with the Nuclear Fuel Complex for supplying gas monitoring systems, demonstrating partial capability in winning government-related procurement. Based on primary research, the company has not disclosed recent tender victories, formal PESO certification details, or compliance metrics for its product catalog.

SUPPORTING POINTS

- Paartha Electronics won a purchase order from the Nuclear Fuel Complex (NFC), Hyderabad, on January 21, 2014, for the supply and installation of Gas Vigil make LPG sensors and a control monitor at the NFC LPG yard.

Local Manufacturing & Partnerships — scored 74/100

> We verified that the firm maintains a small-scale, self-reported local assembly and testing footprint in Telangana to serve regional buyers. Based on primary research, the company has not disclosed strategic joint ventures or technology transfer partnerships with international safety equipment manufacturers.

SUPPORTING POINTS

- The company reports operating a 900-square-foot production facility in Kukatpally, Hyderabad, India, which is equipped with a printed circuit board (PCB) assembly and testing section.
- The company reports supplying its locally produced gas detection systems under the brand Gas Kavach to industrial and domestic clients within India.

Parameter Definitions

X — Product Capability

1. **Sensor Technology & Detection Range**
   Ability to detect a wide range of gases (e.g., LEL, ppm, %vol) with high accuracy and fast response, using advanced sensor types like IR, catalytic, or electrochemical. Evidence includes technical specs, detection limits, and response times.
2. **Product Portfolio Breadth**
   Range of gas and flame detection products offered, including portable detectors, fixed systems, flame detectors, and control panels. Strength shown by number of models, gas types covered, and applications served.
3. **Intrinsic Safety & Certification**
   Compliance with intrinsic safety standards (e.g., ATEX, IECEx, IS) and Indian certifications (PESO, BIS) for hazardous areas. Evidence includes certificates, safety ratings, and approvals for use in explosive environments.
4. **Environmental Durability & Reliability**
   Robustness in extreme conditions (temperature, humidity, dust, corrosion) and long-term reliability with low false alarms. Evidence includes IP ratings, operating temperature ranges, MTBF data, and field reliability reports.
5. **Integration & Smart Features**
   Capability to integrate with control systems (PLC, SCADA, IoT) and offer smart features like wireless connectivity, remote monitoring, and data analytics. Evidence includes communication protocols, software platforms, and automation compatibility.

Y — Business Capability

1. **Installed Base & Key Accounts**
   Number of active installations and major customers in India, such as oil & gas, petrochemical, and manufacturing plants. Evidence includes customer names, project references, and installed unit counts.
2. **Revenue & Growth in India**
   Sales revenue and growth rate in the Indian market, reflecting commercial success. Evidence includes financial reports, year-over-year growth, and market share estimates.
3. **Distribution & Service Network**
   Extent of sales and service coverage across India, including offices, distributors, and service centers. Evidence includes number of locations, partner networks, and service response times.
4. **Regulatory Approvals & Tenders**
   Success in obtaining Indian regulatory approvals and winning government/private tenders. Evidence includes PESO approvals, tender wins, and compliance with local standards.
5. **Local Manufacturing & Partnerships**
   Presence of local manufacturing or strategic partnerships in India for production, assembly, or technology transfer. Evidence includes manufacturing facilities, joint ventures, and local sourcing.

Research Methodology

This quadrant was compiled independently through structured secondary research, supplemented by primary research where available. It is not based on a survey or on vendor self-submission. The steps below describe how companies were identified, screened, evidenced and scored.

1

Market Definition

The market scope and the type of company that qualifies were defined first. Five parameters were then set for each axis, each with a written definition: the X axis measures core capability (products, services or technology) and the Y axis measures business capability.

→

2

Secondary and Primary Research

Secondary research covered company websites, filings, trade directories, industry publications and news. Primary research drew on B2B marketplace listings, where available, to confirm company offerings. Coverage was extended region by region until no new qualifying companies emerged.

→

3

Screening

Each candidate was screened against the market's inclusion criteria and required commercial role. Distributors, resellers, consultancies, contract manufacturers and media names were excluded, and duplicates consolidated. Companies whose main business is an adjacent segment are listed but not placed on the chart.

→

4

Evidence Compilation

Every screened company was assessed on all ten parameters; for chart companies, evidence was recorded parameter by parameter. Claims resting only on a company's own materials are reported as company-reported, partner technology is credited to the partnership, and employee-review sites are not used as customer evidence.

→

5

Scoring

Each parameter is rated 0–100 against its written definition, based on the evidence available. Ratings are analyst assessments under this methodology, not independently audited figures. X and Y are the simple average of the five parameter ratings on each axis; overall is the average of X and Y.

→

6

Normalization & Selection

X and Y scores are rescaled to a 65–100 range across all companies assessed, preserving their rank order. The chart shows an equal number of companies from each of the four quadrants, with boundaries set at the median. All other companies are listed in the table below the chart.

Scoring inputs: each axis is built from five market-specific parameters, defined in the Parameter Definitions section. All parameters are equally weighted.